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Powered stapling device with manual retraction

Abstract

A powered handle assembly includes a motor assembly, a rack, a spur gear, and a manual retract mechanism. The spur gear is movable from a position engaged with the motor assembly and the rack to a positioned disengaged from the motor assembly and engaged with the rack to facilitate manual retraction of the rack.

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6991627	12/2005	Madhani et al.	N/A	N/A
6994714	12/2005	Vargas et al.	N/A	N/A
7000818	12/2005	Shelton, IV et al.	N/A	N/A
7000819	12/2005	Swayze et al.	N/A	N/A
7032799	12/2005	Viola et al.	N/A	N/A
7044352	12/2005	Shelton, IV et al.	N/A	N/A
7044353	12/2005	Mastri et al.	N/A	N/A
7055730	12/2005	Ehrenfels et al.	N/A	N/A
7055731	12/2005	Shelton, IV et al.	N/A	N/A
7059508	12/2005	Shelton, IV et al.	N/A	N/A
7070083	12/2005	Jankowski	N/A	N/A
7083075	12/2005	Swayze et al.	N/A	N/A
7097089	12/2005	Marczyk	N/A	N/A
7111769	12/2005	Wales et al.	N/A	N/A
7114642	12/2005	Whitman	N/A	N/A
7121446	12/2005	Arad et al.	N/A	N/A
7128253	12/2005	Mastri et al.	N/A	N/A
7128254	12/2005	Shelton, IV et al.	N/A	N/A
7140527	12/2005	Ehrenfels et al.	N/A	N/A
7140528	12/2005	Shelton, IV	N/A	N/A
7143923	12/2005	Shelton, IV et al.	N/A	N/A
7143924	12/2005	Scirica et al.	N/A	N/A
7143925	12/2005	Shelton, IV et al.	N/A	N/A
7143926	12/2005	Shelton, IV et al.	N/A	N/A
7147138	12/2005	Shelton, IV	N/A	N/A
7159750	12/2006	Racenet et al.	N/A	N/A
7168604	12/2006	Milliman et al.	N/A	N/A
7172104	12/2006	Scirica et al.	N/A	N/A
7188758	12/2006	Viola et al.	N/A	N/A
7207471	12/2006	Heinrich et al.	N/A	N/A
7213736	12/2006	Wales et al.	N/A	N/A
7225963	12/2006	Scirica	N/A	N/A
7225964	12/2006	Mastri et al.	N/A	N/A
7238195	12/2006	Viola	N/A	N/A
7246734	12/2006	Shelton, IV	N/A	N/A
7258262	12/2006	Mastri et al.	N/A	N/A
7267682	12/2006	Bender et al.	N/A	N/A
7278562	12/2006	Mastri et al.	N/A	N/A
7278563	12/2006	Green	N/A	N/A
7287682	12/2006	Ezzat et al.	N/A	N/A
7293685	12/2006	Ehrenfels et al.	N/A	N/A
7296722	12/2006	Ivanko	N/A	N/A
7296724	12/2006	Green et al.	N/A	N/A
7296772	12/2006	Wang	N/A	N/A
7300444	12/2006	Nielsen et al.	N/A	N/A
7303107	12/2006	Milliman et al.	N/A	N/A
7303108	12/2006	Shelton, IV	N/A	N/A
7308998	12/2006	Mastri et al.	N/A	N/A
7326232	12/2007	Viola et al.	N/A	N/A

7328828	12/2007	Ortiz et al.	N/A	N/A
7328829	12/2007	Arad et al.	N/A	N/A
7334717	12/2007	Rethy et al.	N/A	N/A
7354447	12/2007	Shelton, IV et al.	N/A	N/A
7357287	12/2007	Shelton, IV et al.	N/A	N/A
7364061	12/2007	Swayze et al.	N/A	N/A
7367485	12/2007	Shelton, IV et al.	N/A	N/A
7377928	12/2007	Zubik et al.	N/A	N/A
7380695	12/2007	Doll et al.	N/A	N/A
7380696	12/2007	Shelton, IV et al.	N/A	N/A
7396356	12/2007	Mollenauer	N/A	N/A
7398907	12/2007	Racenet et al.	N/A	N/A
7399310	12/2007	Edoga et al.	N/A	N/A
7401720	12/2007	Durrani	N/A	N/A
7401721	12/2007	Holsten et al.	N/A	N/A
7404508	12/2007	Smith et al.	N/A	N/A
7404509	12/2007	Ortiz et al.	N/A	N/A
7407074	12/2007	Ortiz et al.	N/A	N/A
7407075	12/2007	Holsten et al.	N/A	N/A
7407077	12/2007	Ortiz et al.	N/A	N/A
7407078	12/2007	Shelton, IV et al.	N/A	N/A
7416101	12/2007	Shelton, IV et al.	N/A	N/A
7419080	12/2007	Smith et al.	N/A	N/A
7419081	12/2007	Ehrenfels et al.	N/A	N/A
7419495	12/2007	Menn et al.	N/A	N/A
7422139	12/2007	Shelton, IV et al.	N/A	N/A
7424965	12/2007	Racenet et al.	N/A	N/A
7431189	12/2007	Shelton, IV et al.	N/A	N/A
7431730	12/2007	Viola	N/A	N/A
7434715	12/2007	Shelton, IV et al.	N/A	N/A
7434717	12/2007	Shelton, IV et al.	N/A	N/A
7438208	12/2007	Larson	N/A	N/A
7438209	12/2007	Hess et al.	N/A	N/A
7441684	12/2007	Shelton, IV et al.	N/A	N/A
7441685	12/2007	Boudreaux	N/A	N/A
7448525	12/2007	Shelton, IV et al.	N/A	N/A
7451904	12/2007	Shelton, IV	N/A	N/A
7455208	12/2007	Wales et al.	N/A	N/A
7455676	12/2007	Holsten et al.	N/A	N/A
7458494	12/2007	Matsutani et al.	N/A	N/A
7461767	12/2007	Viola et al.	N/A	N/A
7462185	12/2007	Knodel	N/A	N/A
7464846	12/2007	Shelton, IV et al.	N/A	N/A
7464847	12/2007	Viola et al.	N/A	N/A
7464848	12/2007	Green et al.	N/A	N/A
7464849	12/2007	Shelton, IV et al.	N/A	N/A
7467740	12/2007	Shelton, IV et al.	N/A	N/A
7472814	12/2008	Mastri et al.	N/A	N/A
7472815	12/2008	Shelton, IV et al.	N/A	N/A
7472816	12/2008	Holsten et al.	N/A	N/A

7473258	12/2008	Clauson et al.	N/A	N/A
7481347	12/2008	Roy	N/A	N/A
7481348	12/2008	Marczyk	N/A	N/A
7481349	12/2008	Holsten et al.	N/A	N/A
7481824	12/2008	Boudreaux et al.	N/A	N/A
7487899	12/2008	Shelton, IV et al.	N/A	N/A
7490749	12/2008	Schall et al.	N/A	N/A
7494039	12/2008	Racenet et al.	N/A	N/A
7500979	12/2008	Hueil et al.	N/A	N/A
7503474	12/2008	Hillstead et al.	N/A	N/A
7506790	12/2008	Shelton, IV	N/A	N/A
7506791	12/2008	Omaits et al.	N/A	N/A
7510107	12/2008	Timm et al.	N/A	N/A
7513408	12/2008	Shelton, IV et al.	N/A	N/A
7517356	12/2008	Heinrich	N/A	N/A
7537602	12/2008	Whitman	N/A	N/A
7543729	12/2008	Ivanko	N/A	N/A
7543730	12/2008	Marczyk	N/A	N/A
7543731	12/2008	Green et al.	N/A	N/A
7552854	12/2008	Wixey et al.	N/A	N/A
7556185	12/2008	Viola	N/A	N/A
7556186	12/2008	Milliman	N/A	N/A
7559450	12/2008	Wales et al.	N/A	N/A
7559452	12/2008	Wales et al.	N/A	N/A
7559453	12/2008	Heinrich et al.	N/A	N/A
7559937	12/2008	de la Torre et al.	N/A	N/A
7565993	12/2008	Milliman et al.	N/A	N/A
7568603	12/2008	Shelton, IV et al.	N/A	N/A
7568604	12/2008	Ehrenfels et al.	N/A	N/A
7571845	12/2008	Viola	N/A	N/A
7575144	12/2008	Ortiz et al.	N/A	N/A
7584880	12/2008	Racenet et al.	N/A	N/A
7588174	12/2008	Holsten et al.	N/A	N/A
7588175	12/2008	Timm et al.	N/A	N/A
7588176	12/2008	Timm et al.	N/A	N/A
7588177	12/2008	Racenet	N/A	N/A
7597229	12/2008	Boudreaux et al.	N/A	N/A
7597230	12/2008	Racenet et al.	N/A	N/A
7600663	12/2008	Green	N/A	N/A
7604150	12/2008	Boudreaux	N/A	N/A
7604151	12/2008	Hess et al.	N/A	N/A
7607557	12/2008	Shelton, IV et al.	N/A	N/A
7611038	12/2008	Racenet et al.	N/A	N/A
7617961	12/2008	Viola	N/A	N/A
7624902	12/2008	Marczyk et al.	N/A	N/A
7624903	12/2008	Green et al.	N/A	N/A
7631793	12/2008	Rethy et al.	N/A	N/A
7631794	12/2008	Rethy et al.	N/A	N/A
7635073	12/2008	Heinrich	N/A	N/A
7635074	12/2008	Olson et al.	N/A	N/A

7635373	12/2008	Ortiz	N/A	N/A
7637409	12/2008	Marczyk	N/A	N/A
7637410	12/2008	Marczyk	N/A	N/A
7641091	12/2009	Olson et al.	N/A	N/A
7641095	12/2009	Viola	N/A	N/A
7644848	12/2009	Swayze et al.	N/A	N/A
7648055	12/2009	Marczyk	N/A	N/A
7651017	12/2009	Ortiz et al.	N/A	N/A
7654431	12/2009	Hueil et al.	N/A	N/A
7658311	12/2009	Boudreaux	N/A	N/A
7658312	12/2009	Vidal et al.	N/A	N/A
7665646	12/2009	Prommersberger	N/A	N/A
7665647	12/2009	Shelton, IV et al.	N/A	N/A
7669746	12/2009	Shelton, IV	N/A	N/A
7670334	12/2009	Hueil et al.	N/A	N/A
7673780	12/2009	Shelton, IV et al.	N/A	N/A
7673781	12/2009	Swayze et al.	N/A	N/A
7673782	12/2009	Hess et al.	N/A	N/A
7673783	12/2009	Morgan et al.	N/A	N/A
7678121	12/2009	Knodel	N/A	N/A
7681772	12/2009	Green et al.	N/A	N/A
7682367	12/2009	Shah et al.	N/A	N/A
7682368	12/2009	Bombard et al.	N/A	N/A
7690547	12/2009	Racenet et al.	N/A	N/A
7694865	12/2009	Scirica	N/A	N/A
7699205	12/2009	Ivanko	N/A	N/A
7703653	12/2009	Shah et al.	N/A	N/A
7721931	12/2009	Shelton, IV et al.	N/A	N/A
7721933	12/2009	Ehrenfels et al.	N/A	N/A
7721935	12/2009	Racenet et al.	N/A	N/A
7726537	12/2009	Olson et al.	N/A	N/A
7726538	12/2009	Holsten et al.	N/A	N/A
7726539	12/2009	Holsten et al.	N/A	N/A
7731072	12/2009	Timm et al.	N/A	N/A
7735703	12/2009	Morgan et al.	N/A	N/A
7740159	12/2009	Shelton, IV et al.	N/A	N/A
7740160	12/2009	Viola	N/A	N/A
7743960	12/2009	Whitman et al.	N/A	N/A
7744628	12/2009	Viola	N/A	N/A
7753245	12/2009	Boudreaux et al.	N/A	N/A
7753248	12/2009	Viola	N/A	N/A
7757924	12/2009	Gerbi et al.	N/A	N/A
7757925	12/2009	Viola et al.	N/A	N/A
7762445	12/2009	Heinrich et al.	N/A	N/A
7766209	12/2009	Baxter, III et al.	N/A	N/A
7766210	12/2009	Shelton, IV et al.	N/A	N/A
7766924	12/2009	Bombard et al.	N/A	N/A
7766928	12/2009	Ezzat et al.	N/A	N/A
7770774	12/2009	Mastri et al.	N/A	N/A
7770775	12/2009	Shelton, IV et al.	N/A	N/A

7776060	12/2009	Mooradian et al.	N/A	N/A
7780055	12/2009	Scirica et al.	N/A	N/A
7784662	12/2009	Wales et al.	N/A	N/A
7789283	12/2009	Shah	N/A	N/A
7789889	12/2009	Zubik et al.	N/A	N/A
7793812	12/2009	Moore et al.	N/A	N/A
7793814	12/2009	Racenet et al.	N/A	N/A
7794475	12/2009	Hess et al.	N/A	N/A
7798385	12/2009	Boyden et al.	N/A	N/A
7798386	12/2009	Schall et al.	N/A	N/A
7799039	12/2009	Shelton, IV et al.	N/A	N/A
7810690	12/2009	Bilotti et al.	N/A	N/A
7810692	12/2009	Hall et al.	N/A	N/A
7810693	12/2009	Broehl et al.	N/A	N/A
7815090	12/2009	Marczyk	N/A	N/A
7815091	12/2009	Marczyk	N/A	N/A
7815092	12/2009	Whitman et al.	N/A	N/A
7819296	12/2009	Hueil et al.	N/A	N/A
7819297	12/2009	Doll et al.	N/A	N/A
7819298	12/2009	Hall et al.	N/A	N/A
7819299	12/2009	Shelton, IV et al.	N/A	N/A
7819896	12/2009	Racenet	N/A	N/A
7823760	12/2009	Zemlok et al.	N/A	N/A
7823761	12/2009	Boyden et al.	N/A	N/A
7824426	12/2009	Racenet et al.	N/A	N/A
7828186	12/2009	Wales	N/A	N/A
7828187	12/2009	Green et al.	N/A	N/A
7828188	12/2009	Jankowski	N/A	N/A
7828189	12/2009	Holsten et al.	N/A	N/A
7832408	12/2009	Shelton, IV et al.	N/A	N/A
7832611	12/2009	Boyden et al.	N/A	N/A
7832612	12/2009	Baxter, III et al.	N/A	N/A
7834630	12/2009	Damadian et al.	N/A	N/A
7837079	12/2009	Holsten et al.	N/A	N/A
7837081	12/2009	Holsten et al.	N/A	N/A
7841503	12/2009	Sonnenschein et al.	N/A	N/A
7845533	12/2009	Marczyk et al.	N/A	N/A
7845534	12/2009	Viola et al.	N/A	N/A
7845535	12/2009	Scircia	N/A	N/A
7845537	12/2009	Shelton, IV et al.	N/A	N/A
7845538	12/2009	Whitman	N/A	N/A
7850703	12/2009	Bombard et al.	N/A	N/A
7857183	12/2009	Shelton, IV	N/A	N/A
7857184	12/2009	Viola	N/A	N/A
7857185	12/2009	Swayze et al.	N/A	N/A
7857186	12/2009	Baxter, III et al.	N/A	N/A
7861906	12/2010	Doll et al.	N/A	N/A
7861907	12/2010	Green et al.	N/A	N/A
7866524	12/2010	Krehel	N/A	N/A
7866525	12/2010	Scirica	N/A	N/A

7866526	12/2010	Green et al.	N/A	N/A
7866527	12/2010	Hall et al.	N/A	N/A
7866528	12/2010	Olson et al.	N/A	N/A
7870989	12/2010	Viola et al.	N/A	N/A
7886952	12/2010	Scirica et al.	N/A	N/A
7891532	12/2010	Mastri et al.	N/A	N/A
7891533	12/2010	Green et al.	N/A	N/A
7891534	12/2010	Wenchell et al.	N/A	N/A
7896214	12/2010	Farascioni	N/A	N/A
7900805	12/2010	Shelton, IV et al.	N/A	N/A
7901416	12/2010	Nolan et al.	N/A	N/A
7905380	12/2010	Shelton, IV et al.	N/A	N/A
7905381	12/2010	Baxter, III et al.	N/A	N/A
7909039	12/2010	Hur	N/A	N/A
7909220	12/2010	Viola	N/A	N/A
7909221	12/2010	Viola et al.	N/A	N/A
7909224	12/2010	Prommersberger	N/A	N/A
7913891	12/2010	Doll et al.	N/A	N/A
7913893	12/2010	Mastri et al.	N/A	N/A
7914543	12/2010	Roth et al.	N/A	N/A
7918230	12/2010	Whitman et al.	N/A	N/A
7922061	12/2010	Shelton, IV et al.	N/A	N/A
7922063	12/2010	Zemlok et al.	N/A	N/A
7922064	12/2010	Boyden et al.	N/A	N/A
7926691	12/2010	Viola et al.	N/A	N/A
7926692	12/2010	Racenet et al.	N/A	N/A
7934628	12/2010	Wenchell et al.	N/A	N/A
7934630	12/2010	Shelton, IV et al.	N/A	N/A
7934631	12/2010	Balbierz et al.	N/A	N/A
7942300	12/2010	Rethy et al.	N/A	N/A
7942303	12/2010	Shah	N/A	N/A
7950560	12/2010	Zemlok et al.	N/A	N/A
7950561	12/2010	Aranyi	N/A	N/A
7950562	12/2010	Beardsley et al.	N/A	N/A
7954682	12/2010	Giordano et al.	N/A	N/A
7954683	12/2010	Knodel et al.	N/A	N/A
7954684	12/2010	Boudreaux	N/A	N/A
7954685	12/2010	Viola	N/A	N/A
7954686	12/2010	Baxter, III et al.	N/A	N/A
7954687	12/2010	Zemlok et al.	N/A	N/A
7959051	12/2010	Smith et al.	N/A	N/A
7963431	12/2010	Scirica	N/A	N/A
7963432	12/2010	Knodel et al.	N/A	N/A
7963433	12/2010	Whitman et al.	N/A	N/A
7967178	12/2010	Scirica et al.	N/A	N/A
7967179	12/2010	Olson et al.	N/A	N/A
7967180	12/2010	Scirica	N/A	N/A
7975894	12/2010	Boyden et al.	N/A	N/A
7980443	12/2010	Scheib et al.	N/A	N/A
7988026	12/2010	Knodel et al.	N/A	N/A

7988027	12/2010	Olson et al.	N/A	N/A
7988028	12/2010	Farascioni et al.	N/A	N/A
7992758	12/2010	Whitman et al.	N/A	N/A
7997468	12/2010	Farascioni	N/A	N/A
7997469	12/2010	Olson et al.	N/A	N/A
8002795	12/2010	Beetel	N/A	N/A
8006885	12/2010	Marczyk	N/A	N/A
8006887	12/2010	Marczyk	N/A	N/A
8007505	12/2010	Weller et al.	N/A	N/A
8007513	12/2010	Nalagatla et al.	N/A	N/A
8011550	12/2010	Aranyi et al.	N/A	N/A
8011551	12/2010	Marczyk et al.	N/A	N/A
8011552	12/2010	Ivanko	N/A	N/A
8011553	12/2010	Mastri et al.	N/A	N/A
8011555	12/2010	Tarinelli et al.	N/A	N/A
8012170	12/2010	Whitman et al.	N/A	N/A
8015976	12/2010	Shah	N/A	N/A
8016177	12/2010	Bettuchi et al.	N/A	N/A
8016178	12/2010	Olson et al.	N/A	N/A
8020742	12/2010	Marczyk	N/A	N/A
8020743	12/2010	Shelton, IV	N/A	N/A
8028882	12/2010	Viola	N/A	N/A
8028883	12/2010	Stopek	N/A	N/A
8028884	12/2010	Sniffin et al.	N/A	N/A
8033438	12/2010	Scirica	N/A	N/A
8033440	12/2010	Wenchell et al.	N/A	N/A
8033441	12/2010	Marczyk	N/A	N/A
8033442	12/2010	Racenet et al.	N/A	N/A
8034077	12/2010	Smith et al.	N/A	N/A
8038044	12/2010	Viola	N/A	N/A
8038045	12/2010	Bettuchi et al.	N/A	N/A
8052024	12/2010	Viola et al.	N/A	N/A
8056787	12/2010	Boudreaux et al.	N/A	N/A
8056788	12/2010	Mastri et al.	N/A	N/A
8056791	12/2010	Whitman	N/A	N/A
8061577	12/2010	Racenet et al.	N/A	N/A
8066166	12/2010	Demmy et al.	N/A	N/A
8070033	12/2010	Milliman et al.	N/A	N/A
8070034	12/2010	Knodel	N/A	N/A
8070035	12/2010	Holsten et al.	N/A	N/A
8074858	12/2010	Marczyk	N/A	N/A
8074859	12/2010	Kostrzewski	N/A	N/A
8074862	12/2010	Shah	N/A	N/A
8083118	12/2010	Milliman et al.	N/A	N/A
8083119	12/2010	Prommersberger	N/A	N/A
8083120	12/2010	Shelton, IV et al.	N/A	N/A
8087563	12/2011	Milliman et al.	N/A	N/A
8091753	12/2011	Viola	N/A	N/A
8091754	12/2011	Ehrenfels et al.	N/A	N/A
8091756	12/2011	Viola	N/A	N/A

8092493	12/2011	Marczyk	N/A	N/A
8096459	12/2011	Ortiz et al.	N/A	N/A
8096460	12/2011	Blier et al.	N/A	N/A
8100309	12/2011	Marczyk	N/A	N/A
8100310	12/2011	Zemlok	N/A	N/A
8102008	12/2011	Wells	N/A	N/A
8113406	12/2011	Holsten et al.	N/A	N/A
8113407	12/2011	Holsten et al.	N/A	N/A
8113408	12/2011	Wenchell et al.	N/A	N/A
8113409	12/2011	Cohen et al.	N/A	N/A
8113410	12/2011	Hall et al.	N/A	N/A
8123101	12/2011	Racenet et al.	N/A	N/A
8127975	12/2011	Olson et al.	N/A	N/A
8127976	12/2011	Scirica et al.	N/A	N/A
8132703	12/2011	Milliman et al.	N/A	N/A
8132705	12/2011	Viola et al.	N/A	N/A
8132706	12/2011	Marczyk et al.	N/A	N/A
8136713	12/2011	Hathaway et al.	N/A	N/A
8141762	12/2011	Bedi et al.	N/A	N/A
8152041	12/2011	Kostrzewski	N/A	N/A
8157148	12/2011	Scirica	N/A	N/A
8157150	12/2011	Viola et al.	N/A	N/A
8157151	12/2011	Ingmanson et al.	N/A	N/A
8157152	12/2011	Holsten et al.	N/A	N/A
8162197	12/2011	Mastri et al.	N/A	N/A
8167185	12/2011	Shelton, IV et al.	N/A	N/A
8167186	12/2011	Racenet et al.	N/A	N/A
8172121	12/2011	Krehel	N/A	N/A
8172124	12/2011	Shelton, IV et al.	N/A	N/A
8181837	12/2011	Roy	N/A	N/A
8186555	12/2011	Shelton, IV et al.	N/A	N/A
8186557	12/2011	Cohen et al.	N/A	N/A
8186558	12/2011	Sapienza	N/A	N/A
8186559	12/2011	Whitman	N/A	N/A
8186560	12/2011	Hess et al.	N/A	N/A
8193044	12/2011	Kenneth	N/A	N/A
8196795	12/2011	Moore et al.	N/A	N/A
8196796	12/2011	Shelton, IV et al.	N/A	N/A
8201721	12/2011	Zemlok et al.	N/A	N/A
8205619	12/2011	Shah et al.	N/A	N/A
8205780	12/2011	Sorrentino et al.	N/A	N/A
8205781	12/2011	Baxter, III et al.	N/A	N/A
8210412	12/2011	Marczyk	N/A	N/A
8210416	12/2011	Milliman et al.	N/A	N/A
8215532	12/2011	Marczyk	N/A	N/A
8216236	12/2011	Heinrich et al.	N/A	N/A
8220688	12/2011	Laurent et al.	N/A	N/A
8220690	12/2011	Hess et al.	N/A	N/A
8225979	12/2011	Farascioni et al.	N/A	N/A
8231040	12/2011	Zemlok et al.	N/A	N/A

8231041	12/2011	Marczyk et al.	N/A	N/A
8235272	12/2011	Nicholas et al.	N/A	N/A
8235273	12/2011	Olson et al.	N/A	N/A
8235274	12/2011	Cappola	N/A	N/A
8236010	12/2011	Ortiz et al.	N/A	N/A
8240536	12/2011	Marczyk	N/A	N/A
8240537	12/2011	Marczyk	N/A	N/A
8241322	12/2011	Whitman et al.	N/A	N/A
8245897	12/2011	Tzakis et al.	N/A	N/A
8245898	12/2011	Smith et al.	N/A	N/A
8245899	12/2011	Swensgard et al.	N/A	N/A
8245931	12/2011	Shigeta	N/A	N/A
8252009	12/2011	Weller et al.	N/A	N/A
8256653	12/2011	Farascioni	N/A	N/A
8256654	12/2011	Bettuchi et al.	N/A	N/A
8256655	12/2011	Sniffin et al.	N/A	N/A
8256656	12/2011	Milliman et al.	N/A	N/A
8267300	12/2011	Boudreaux	N/A	N/A
8272551	12/2011	Knodel et al.	N/A	N/A
8272553	12/2011	Mastri et al.	N/A	N/A
8272554	12/2011	Whitman et al.	N/A	N/A
8276594	12/2011	Shah	N/A	N/A
8276801	12/2011	Zemlok et al.	N/A	N/A
8281973	12/2011	Wenchell et al.	N/A	N/A
8286847	12/2011	Taylor	N/A	N/A
8286848	12/2011	Wenchell et al.	N/A	N/A
8286850	12/2011	Viola	N/A	N/A
8292146	12/2011	Holsten et al.	N/A	N/A
8292147	12/2011	Viola	N/A	N/A
8292148	12/2011	Viola	N/A	N/A
8292149	12/2011	Ivanko	N/A	N/A
8292150	12/2011	Bryant	N/A	N/A
8292151	12/2011	Viola	N/A	N/A
8292152	12/2011	Milliman et al.	N/A	N/A
8292153	12/2011	Jankowski	N/A	N/A
8292154	12/2011	Marczyk	N/A	N/A
8292155	12/2011	Shelton, IV et al.	N/A	N/A
8292156	12/2011	Kostrzewski	N/A	N/A
8292158	12/2011	Sapienza	N/A	N/A
8308040	12/2011	Huang et al.	N/A	N/A
8308041	12/2011	Kostrzewski	N/A	N/A
8308042	12/2011	Aranyi	N/A	N/A
8308043	12/2011	Bindra et al.	N/A	N/A
8308044	12/2011	Viola	N/A	N/A
8308046	12/2011	Prommersberger	N/A	N/A
8308757	12/2011	Hillstead et al.	N/A	N/A
8317070	12/2011	Hueil et al.	N/A	N/A
8317071	12/2011	Knodel	N/A	N/A
8322455	12/2011	Shelton, IV et al.	N/A	N/A
8322589	12/2011	Boudreaux	N/A	N/A

8328061	12/2011	Kasvikis	N/A	N/A
8328065	12/2011	Shah	N/A	N/A
8333313	12/2011	Boudreaux et al.	N/A	N/A
8336751	12/2011	Scirica	N/A	N/A
8336753	12/2011	Olson et al.	N/A	N/A
8336754	12/2011	Cappola et al.	N/A	N/A
8342377	12/2012	Milliman et al.	N/A	N/A
8342378	12/2012	Marczyk et al.	N/A	N/A
8342379	12/2012	Whitman et al.	N/A	N/A
8342380	12/2012	Viola	N/A	N/A
8348123	12/2012	Scirica et al.	N/A	N/A
8348124	12/2012	Scirica	N/A	N/A
8348125	12/2012	Viola et al.	N/A	N/A
8348126	12/2012	Olson et al.	N/A	N/A
8348127	12/2012	Marczyk	N/A	N/A
8348129	12/2012	Bedi et al.	N/A	N/A
8348130	12/2012	Shah et al.	N/A	N/A
8348131	12/2012	Omaits et al.	N/A	N/A
8353437	12/2012	Boudreaux	N/A	N/A
8353440	12/2012	Whitman et al.	N/A	N/A
8356740	12/2012	Knodel	N/A	N/A
8357174	12/2012	Roth et al.	N/A	N/A
8360294	12/2012	Scirica	N/A	N/A
8360297	12/2012	Shelton, IV et al.	N/A	N/A
8360298	12/2012	Farascioni et al.	N/A	N/A
8360299	12/2012	Zemlok et al.	N/A	N/A
8365971	12/2012	Knodel	N/A	N/A
8365972	12/2012	Aranyi et al.	N/A	N/A
8365973	12/2012	White et al.	N/A	N/A
8365976	12/2012	Hess et al.	N/A	N/A
8371491	12/2012	Huitema et al.	N/A	N/A
8371492	12/2012	Aranyi et al.	N/A	N/A
8371493	12/2012	Aranyi et al.	N/A	N/A
8381828	12/2012	Whitman et al.	N/A	N/A
8381961	12/2012	Holsten et al.	N/A	N/A
8387848	12/2012	Johnson et al.	N/A	N/A
8387849	12/2012	Buesseler et al.	N/A	N/A
8387850	12/2012	Hathaway et al.	N/A	N/A
8388652	12/2012	Viola	N/A	N/A
8393513	12/2012	Jankowski	N/A	N/A
8393514	12/2012	Shelton, IV et al.	N/A	N/A
8393516	12/2012	Kostrzewski	N/A	N/A
8397971	12/2012	Yates et al.	N/A	N/A
8397972	12/2012	Kostrzewski	N/A	N/A
8403195	12/2012	Beardsley et al.	N/A	N/A
8403196	12/2012	Beardsley et al.	N/A	N/A
8403197	12/2012	Vidal et al.	N/A	N/A
8403198	12/2012	Sorrentino et al.	N/A	N/A
8403956	12/2012	Thompson et al.	N/A	N/A
8408439	12/2012	Huang et al.	N/A	N/A

8408440	12/2012	Olson et al.	N/A	N/A
8408442	12/2012	Racenet et al.	N/A	N/A
8413868	12/2012	Cappola	N/A	N/A
8413869	12/2012	Heinrich	N/A	N/A
8413871	12/2012	Racenet et al.	N/A	N/A
8418904	12/2012	Wenchell et al.	N/A	N/A
8418905	12/2012	Milliman	N/A	N/A
8418906	12/2012	Farascioni et al.	N/A	N/A
8418907	12/2012	Johnson et al.	N/A	N/A
8418908	12/2012	Beardsley	N/A	N/A
8419768	12/2012	Marczyk	N/A	N/A
8424735	12/2012	Viola et al.	N/A	N/A
8424736	12/2012	Scirica et al.	N/A	N/A
8424737	12/2012	Scirica	N/A	N/A
8424739	12/2012	Racenet et al.	N/A	N/A
8424740	12/2012	Shelton, IV et al.	N/A	N/A
8439244	12/2012	Holcomb et al.	N/A	N/A
8439245	12/2012	Knodel et al.	N/A	N/A
8439246	12/2012	Knodel	N/A	N/A
8444036	12/2012	Shelton, IV	N/A	N/A
8444037	12/2012	Nicholas et al.	N/A	N/A
8444038	12/2012	Farascioni et al.	N/A	N/A
8448832	12/2012	Viola et al.	N/A	N/A
8453652	12/2012	Stopek	N/A	N/A
8453905	12/2012	Holcomb et al.	N/A	N/A
8453906	12/2012	Huang et al.	N/A	N/A
8453907	12/2012	Laurent et al.	N/A	N/A
8453908	12/2012	Bedi et al.	N/A	N/A
8453909	12/2012	Olson et al.	N/A	N/A
8453910	12/2012	Bettuchi et al.	N/A	N/A
8453912	12/2012	Mastri et al.	N/A	N/A
8453913	12/2012	Milliman	N/A	N/A
8453914	12/2012	Laurent et al.	N/A	N/A
8454628	12/2012	Smith et al.	N/A	N/A
8459520	12/2012	Giordano et al.	N/A	N/A
8459521	12/2012	Zemlok et al.	N/A	N/A
8459522	12/2012	Marczyk	N/A	N/A
8459523	12/2012	Whitman	N/A	N/A
8459524	12/2012	Pribanic et al.	N/A	N/A
8459525	12/2012	Yates et al.	N/A	N/A
8464922	12/2012	Marczyk	N/A	N/A
8464923	12/2012	Shelton, IV	N/A	N/A
8469252	12/2012	Holcomb et al.	N/A	N/A
8469254	12/2012	Czernik et al.	N/A	N/A
8474677	12/2012	Woodard, Jr. et al.	N/A	N/A
8479967	12/2012	Marczyk	N/A	N/A
8479968	12/2012	Hodgkinson et al.	N/A	N/A
8479969	12/2012	Shelton, IV	N/A	N/A
8485412	12/2012	Shelton, IV et al.	N/A	N/A
8490852	12/2012	Viola	N/A	N/A

8496152	12/2012	Viola	N/A	N/A
8496154	12/2012	Marczyk et al.	N/A	N/A
8496156	12/2012	Sniffin et al.	N/A	N/A
8496683	12/2012	Prommersberger et al.	N/A	N/A
8499993	12/2012	Shelton, IV et al.	N/A	N/A
8505799	12/2012	Viola et al.	N/A	N/A
8505802	12/2012	Viola et al.	N/A	N/A
8511575	12/2012	Cok	N/A	N/A
8512359	12/2012	Whitman et al.	N/A	N/A
8512402	12/2012	Marczyk et al.	N/A	N/A
8517240	12/2012	Mata et al.	N/A	N/A
8517241	12/2012	Nicholas et al.	N/A	N/A
8517243	12/2012	Giordano et al.	N/A	N/A
8517244	12/2012	Shelton, IV et al.	N/A	N/A
8523041	12/2012	Ishitsuki et al.	N/A	N/A
8523042	12/2012	Masiakos et al.	N/A	N/A
8523043	12/2012	Ullrich et al.	N/A	N/A
8534528	12/2012	Shelton, IV	N/A	N/A
8540128	12/2012	Shelton, IV et al.	N/A	N/A
8540129	12/2012	Baxter, III et al.	N/A	N/A
8540130	12/2012	Moore et al.	N/A	N/A
8540131	12/2012	Swayze	N/A	N/A
8540733	12/2012	Whitman et al.	N/A	N/A
8544711	12/2012	Ma et al.	N/A	N/A
8550325	12/2012	Cohen et al.	N/A	N/A
8556151	12/2012	Viola	N/A	N/A
8561870	12/2012	Baxter, III et al.	N/A	N/A
8561873	12/2012	Ingmanson et al.	N/A	N/A
8561874	12/2012	Scirica	N/A	N/A
8567656	12/2012	Shelton, IV et al.	N/A	N/A
8573461	12/2012	Shelton, IV et al.	N/A	N/A
8573463	12/2012	Scirica et al.	N/A	N/A
8573465	12/2012	Shelton, IV	N/A	N/A
8579176	12/2012	Smith et al.	N/A	N/A
8579177	12/2012	Beetel	N/A	N/A
8584919	12/2012	Hueil et al.	N/A	N/A
8584920	12/2012	Hodgkinson	N/A	N/A
8590762	12/2012	Hess et al.	N/A	N/A
8596515	12/2012	Okoniewski	N/A	N/A
8597311	12/2012	Criscuolo et al.	N/A	N/A
8602288	12/2012	Shelton, IV et al.	N/A	N/A
8608045	12/2012	Smith et al.	N/A	N/A
8608046	12/2012	Laurent et al.	N/A	N/A
8608047	12/2012	Holsten et al.	N/A	N/A
8613383	12/2012	Beckman et al.	N/A	N/A
8613384	12/2012	Pastorelli et al.	N/A	N/A
8616427	12/2012	Viola	N/A	N/A
8616430	12/2012	Stopek et al.	N/A	N/A
8627994	12/2013	Zemlok et al.	N/A	N/A

8628544	12/2013	Farascioni	N/A	N/A
8631988	12/2013	Viola	N/A	N/A
8631989	12/2013	Aranyi et al.	N/A	N/A
8631991	12/2013	Cropper et al.	N/A	N/A
8632525	12/2013	Kerr et al.	N/A	N/A
8632535	12/2013	Shelton, IV et al.	N/A	N/A
8636187	12/2013	Hueil et al.	N/A	N/A
8636190	12/2013	Zemlok et al.	N/A	N/A
8636192	12/2013	Farascioni et al.	N/A	N/A
8636762	12/2013	Whitman et al.	N/A	N/A
8636766	12/2013	Milliman et al.	N/A	N/A
8640940	12/2013	Ohdaira	N/A	N/A
8657174	12/2013	Yates et al.	N/A	N/A
8657177	12/2013	Scirica et al.	N/A	N/A
8657178	12/2013	Hueil et al.	N/A	N/A
8662371	12/2013	Viola	N/A	N/A
8668129	12/2013	Olson	N/A	N/A
8672206	12/2013	Aranyi et al.	N/A	N/A
8672208	12/2013	Hess et al.	N/A	N/A
8672209	12/2013	Crainich	N/A	N/A
8678263	12/2013	Viola	N/A	N/A
8678990	12/2013	Wazer et al.	N/A	N/A
8679155	12/2013	Knodel et al.	N/A	N/A
8684247	12/2013	Scirica et al.	N/A	N/A
8684249	12/2013	Racenet et al.	N/A	N/A
8684253	12/2013	Giordano et al.	N/A	N/A
8690039	12/2013	Beardsley et al.	N/A	N/A
8695865	12/2013	Smith et al.	N/A	N/A
8695866	12/2013	Leimbach et al.	N/A	N/A
8701958	12/2013	Shelton, IV et al.	N/A	N/A
8701959	12/2013	Shah	N/A	N/A
8701961	12/2013	Ivanko	N/A	N/A
8708213	12/2013	Shelton, IV et al.	N/A	N/A
8714429	12/2013	Demmy	N/A	N/A
8715277	12/2013	Weizman	N/A	N/A
8720766	12/2013	Hess et al.	N/A	N/A
8721630	12/2013	Ortiz et al.	N/A	N/A
8727197	12/2013	Hess et al.	N/A	N/A
8727200	12/2013	Roy	N/A	N/A
8733612	12/2013	Ma	N/A	N/A
8740034	12/2013	Morgan et al.	N/A	N/A
8740039	12/2013	Farascioni	N/A	N/A
8746529	12/2013	Shelton, IV et al.	N/A	N/A
8746530	12/2013	Giordano et al.	N/A	N/A
8746535	12/2013	Shelton, IV et al.	N/A	N/A
8752748	12/2013	Whitman et al.	N/A	N/A
8752749	12/2013	Moore et al.	N/A	N/A
8757465	12/2013	Woodard, Jr. et al.	N/A	N/A
8758391	12/2013	Swayze et al.	N/A	N/A
8763875	12/2013	Morgan et al.	N/A	N/A

8763876	12/2013	Kostrzewski	N/A	N/A
8763877	12/2013	Schall et al.	N/A	N/A
8763879	12/2013	Shelton, IV et al.	N/A	N/A
8770458	12/2013	Scirica	N/A	N/A
8777082	12/2013	Scirica	N/A	N/A
8783541	12/2013	Shelton, IV et al.	N/A	N/A
8783542	12/2013	Riestenberg et al.	N/A	N/A
8789737	12/2013	Hodgkinson et al.	N/A	N/A
8789738	12/2013	Knodel et al.	N/A	N/A
8789739	12/2013	Swensgard	N/A	N/A
8800838	12/2013	Shelton, IV	N/A	N/A
8800840	12/2013	Jankowski	N/A	N/A
8800841	12/2013	Ellerhorst et al.	N/A	N/A
8808311	12/2013	Heinrich et al.	N/A	N/A
8814024	12/2013	Woodard, Jr. et al.	N/A	N/A
8814025	12/2013	Miller et al.	N/A	N/A
8820603	12/2013	Shelton, IV et al.	N/A	N/A
8820605	12/2013	Shelton, IV	N/A	N/A
8820606	12/2013	Hodgkinson	N/A	N/A
8820607	12/2013	Marczyk	N/A	N/A
8827133	12/2013	Shelton, IV et al.	N/A	N/A
8827134	12/2013	Viola et al.	N/A	N/A
8833631	12/2013	Munro, III et al.	N/A	N/A
8833632	12/2013	Swensgard	N/A	N/A
8840003	12/2013	Morgan et al.	N/A	N/A
8840603	12/2013	Shelton, IV et al.	N/A	N/A
8844788	12/2013	Knodel	N/A	N/A
8851354	12/2013	Swensgard et al.	N/A	N/A
8851355	12/2013	Aranyi et al.	N/A	N/A
8857693	12/2013	Schuckmann et al.	N/A	N/A
8864007	12/2013	Widenhouse et al.	N/A	N/A
8864009	12/2013	Shelton, IV et al.	N/A	N/A
8864010	12/2013	Williams	N/A	N/A
8870050	12/2013	Hodgkinson	N/A	N/A
8875971	12/2013	Hall et al.	N/A	N/A
8875972	12/2013	Weisenburgh, II et al.	N/A	N/A
8893949	12/2013	Shelton, IV et al.	N/A	N/A
8893950	12/2013	Marczyk	N/A	N/A
8899461	12/2013	Farascioni	N/A	N/A
8899462	12/2013	Kostrzewski et al.	N/A	N/A
8899463	12/2013	Schall et al.	N/A	N/A
8899464	12/2013	Hueil et al.	N/A	N/A
8900616	12/2013	Belcheva et al.	N/A	N/A
8920435	12/2013	Smith et al.	N/A	N/A
8925782	12/2014	Shelton, IV	N/A	N/A
8926598	12/2014	Mollere et al.	N/A	N/A
8931681	12/2014	Kostrzewski	N/A	N/A
8931682	12/2014	Timm et al.	N/A	N/A
8931693	12/2014	Kumar et al.	N/A	N/A
8955732	12/2014	Zemlok et al.	N/A	N/A

8958429	12/2014	Shukla et al.	N/A	N/A
8960517	12/2014	Lee	N/A	N/A
8967443	12/2014	McCuen	N/A	N/A
8967446	12/2014	Beardsley et al.	N/A	N/A
8973803	12/2014	Hall et al.	N/A	N/A
8978954	12/2014	Shelton, IV et al.	N/A	N/A
8978956	12/2014	Schall et al.	N/A	N/A
8991677	12/2014	Moore et al.	N/A	N/A
8991678	12/2014	Wellman et al.	N/A	N/A
8998058	12/2014	Moore et al.	N/A	N/A
8998060	12/2014	Bruewer et al.	N/A	N/A
9005230	12/2014	Yates et al.	N/A	N/A
9010606	12/2014	Aranyi et al.	N/A	N/A
9010607	12/2014	Kostrzewski	N/A	N/A
9010610	12/2014	Hodgkinson	N/A	N/A
9016539	12/2014	Kostrzewski et al.	N/A	N/A
9016541	12/2014	Viola et al.	N/A	N/A
9016542	12/2014	Shelton, IV et al.	N/A	N/A
9016544	12/2014	Hodgkinson et al.	N/A	N/A
9016545	12/2014	Aranyi et al.	N/A	N/A
9016546	12/2014	Demmy et al.	N/A	N/A
9022271	12/2014	Scirica	N/A	N/A
9027817	12/2014	Milliman et al.	N/A	N/A
9033203	12/2014	Woodard, Jr. et al.	N/A	N/A
9039557	12/2014	Naughton	221/229	B05C 17/0103
9044228	12/2014	Woodard, Jr. et al.	N/A	N/A
9044229	12/2014	Scheib et al.	N/A	N/A
9050084	12/2014	Schmid et al.	N/A	N/A
9055941	12/2014	Schmid et al.	N/A	N/A
9060770	12/2014	Shelton, IV et al.	N/A	N/A
9072535	12/2014	Shelton, IV et al.	N/A	N/A
9084601	12/2014	Moore et al.	N/A	N/A
9089326	12/2014	Krumanaker et al.	N/A	N/A
9095339	12/2014	Moore et al.	N/A	N/A
9101359	12/2014	Smith et al.	N/A	N/A
9107663	12/2014	Swensgard	N/A	N/A
9107664	12/2014	Marczyk	N/A	N/A
9107667	12/2014	Hodgkinson	N/A	N/A
9113862	12/2014	Morgan et al.	N/A	N/A
9113864	12/2014	Morgan et al.	N/A	N/A
9113870	12/2014	Viola	N/A	N/A
9113872	12/2014	Viola	N/A	N/A
9113880	12/2014	Zemlok et al.	N/A	N/A
9125649	12/2014	Bruewer et al.	N/A	N/A
9138225	12/2014	Huang et al.	N/A	N/A
9155537	12/2014	Katre et al.	N/A	N/A
9179912	12/2014	Yates et al.	N/A	N/A
9192378	12/2014	Aranyi et al.	N/A	N/A
9192379	12/2014	Aranyi et al.	N/A	N/A

9192384	12/2014	Bettuchi	N/A	N/A
9198644	12/2014	Balek et al.	N/A	N/A
9198660	12/2014	Hodgkinson	N/A	N/A
9198661	12/2014	Swensgard	N/A	N/A
9198662	12/2014	Barton et al.	N/A	N/A
9204876	12/2014	Cappola et al.	N/A	N/A
9204880	12/2014	Baxter, III et al.	N/A	N/A
9216019	12/2014	Schmid et al.	N/A	N/A
9216020	12/2014	Zhang et al.	N/A	N/A
9220500	12/2014	Swayze et al.	N/A	N/A
9220501	12/2014	Baxter, III et al.	N/A	N/A
9220502	12/2014	Zemlok et al.	N/A	N/A
9232941	12/2015	Mandakolathur Vasudevan et al.	N/A	N/A
9232944	12/2015	Cappola et al.	N/A	N/A
9237891	12/2015	Shelton, IV	N/A	N/A
9237892	12/2015	Hodgkinson	N/A	N/A
9254180	12/2015	Huitema et al.	N/A	N/A
9265585	12/2015	Wingardner et al.	N/A	N/A
9271728	12/2015	Gupta et al.	N/A	N/A
9272406	12/2015	Aronhalt et al.	N/A	N/A
9277919	12/2015	Timmer et al.	N/A	N/A
9282962	12/2015	Schmid et al.	N/A	N/A
9283054	12/2015	Morgan et al.	N/A	N/A
9289209	12/2015	Gurumurthy et al.	N/A	N/A
9289210	12/2015	Baxter, III et al.	N/A	N/A
9289225	12/2015	Shelton, IV et al.	N/A	N/A
9295464	12/2015	Shelton, IV et al.	N/A	N/A
9295465	12/2015	Farascioni	N/A	N/A
9301752	12/2015	Mandakolathur Vasudevan et al.	N/A	N/A
9301753	12/2015	Aldridge et al.	N/A	N/A
9301757	12/2015	Williams	N/A	N/A
9307965	12/2015	Ming et al.	N/A	N/A
9307986	12/2015	Hall et al.	N/A	N/A
9307989	12/2015	Shelton, IV et al.	N/A	N/A
9314246	12/2015	Shelton, IV et al.	N/A	N/A
9320518	12/2015	Henderson et al.	N/A	N/A
9320521	12/2015	Shelton, IV et al.	N/A	N/A
9320523	12/2015	Shelton, IV et al.	N/A	N/A
9326767	12/2015	Koch, Jr. et al.	N/A	N/A
9332987	12/2015	Leimbach et al.	N/A	N/A
9345477	12/2015	Anim et al.	N/A	N/A
9345478	12/2015	Knodel	N/A	N/A
9345481	12/2015	Hall et al.	N/A	N/A
9345780	12/2015	Manoharan et al.	N/A	N/A
9351727	12/2015	Leimbach et al.	N/A	N/A
9351732	12/2015	Hodgkinson	N/A	N/A
9358003	12/2015	Hall et al.	N/A	N/A
9364217	12/2015	Kostrzewski et al.	N/A	N/A

9364218	12/2015	Scirica	N/A	N/A
9364219	12/2015	Olson et al.	N/A	N/A
9364220	12/2015	Williams	N/A	N/A
9364227	12/2015	Kostrzewski	N/A	N/A
9364231	12/2015	Wenchell	N/A	N/A
9364233	12/2015	Alexander, III et al.	N/A	N/A
9370358	12/2015	Shelton, IV et al.	N/A	N/A
9370362	12/2015	Petty et al.	N/A	N/A
9386983	12/2015	Swensgard et al.	N/A	N/A
9386984	12/2015	Aronhalt et al.	N/A	N/A
9386988	12/2015	Baxter, III et al.	N/A	N/A
9393018	12/2015	Wang et al.	N/A	N/A
9398911	12/2015	Auld	N/A	N/A
9402604	12/2015	Williams et al.	N/A	N/A
9421014	12/2015	Ingmanson et al.	N/A	N/A
9433419	12/2015	Gonzalez et al.	N/A	N/A
9433420	12/2015	Hodgkinson	N/A	N/A
9445810	12/2015	Cappola	N/A	N/A
9445813	12/2015	Shelton, IV et al.	N/A	N/A
9451959	12/2015	Patankar et al.	N/A	N/A
9468438	12/2015	Baber et al.	N/A	N/A
9468439	12/2015	Cappola et al.	N/A	N/A
9480476	12/2015	Aldridge et al.	N/A	N/A
9480492	12/2015	Aranyi et al.	N/A	N/A
9492171	12/2015	Patenaude	N/A	N/A
9498212	12/2015	Racenet et al.	N/A	N/A
9498219	12/2015	Moore et al.	N/A	N/A
9510827	12/2015	Kostrzewski	N/A	N/A
9517065	12/2015	Simms et al.	N/A	N/A
9517066	12/2015	Racenet et al.	N/A	N/A
9539007	12/2016	Dhakad et al.	N/A	N/A
9549735	12/2016	Shelton, IV et al.	N/A	N/A
9561031	12/2016	Heinrich et al.	N/A	N/A
9597077	12/2016	Hodgkinson	N/A	N/A
9610080	12/2016	Whitfield et al.	N/A	N/A
9913646	12/2017	Shelton, IV	N/A	N/A
9936954	12/2017	Shelton, IV et al.	N/A	N/A
9987099	12/2017	Chen et al.	N/A	N/A
10004497	12/2017	Overmyer et al.	N/A	N/A
10004505	12/2017	Moore et al.	N/A	N/A
10045782	12/2017	Murthy Aravalli	N/A	N/A
10064622	12/2017	Murthy Aravalli	N/A	N/A
10076325	12/2017	Huang et al.	N/A	N/A
10135242	12/2017	Baber et al.	N/A	N/A
10143474	12/2017	Bucciaglia et al.	N/A	N/A
10172615	12/2018	Marczyk et al.	N/A	N/A
10206677	12/2018	Harris et al.	N/A	N/A
10265066	12/2018	Measamer et al.	N/A	N/A
10265074	12/2018	Shelton, IV et al.	N/A	N/A
10271841	12/2018	Overmyer et al.	N/A	N/A

10327776	12/2018	Harris et al.	N/A	N/A
10349941	12/2018	Marczyk et al.	N/A	N/A
10390828	12/2018	Vendely et al.	N/A	N/A
10405857	12/2018	Shelton	N/A	N/A
10463368	12/2018	Kostrzewski	N/A	N/A
10478182	12/2018	Taylor	N/A	N/A
10478183	12/2018	Hess et al.	N/A	N/A
10485541	12/2018	Shelton, IV	N/A	A61B 17/068
10548599	12/2019	Marczyk et al.	N/A	N/A
11771423	12/2022	Chen et al.	N/A	N/A
2004/0108357	12/2003	Milliman et al.	N/A	N/A
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2008/0029573	12/2007	Shelton et al.	N/A	N/A
2008/0029574	12/2007	Shelton et al.	N/A	N/A
2008/0029575	12/2007	Shelton et al.	N/A	N/A
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2008/0110961	12/2007	Voegele et al.	N/A	N/A
2008/0169328	12/2007	Shelton	N/A	N/A
2008/0169332	12/2007	Shelton et al.	N/A	N/A
2008/0169333	12/2007	Shelton et al.	N/A	N/A
2008/0287987	12/2007	Boyden et al.	N/A	N/A
2008/0296346	12/2007	Shelton, IV et al.	N/A	N/A
2008/0308602	12/2007	Timm et al.	N/A	N/A
2008/0308603	12/2007	Shelton et al.	N/A	N/A
2009/0001121	12/2008	Hess et al.	N/A	N/A
2009/0001130	12/2008	Hess et al.	N/A	N/A
2009/0090763	12/2008	Zemlok et al.	N/A	N/A

2009/0090766	12/2008	Knodel	N/A	N/A
2009/0242610	12/2008	Shelton, IV et al.	N/A	N/A
2009/0255974	12/2008	Viola	N/A	N/A
2009/0308907	12/2008	Nalagatla et al.	N/A	N/A
2010/0012703	12/2009	Calabrese et al.	N/A	N/A
2010/0051669	12/2009	Milliman	N/A	N/A
2010/0069942	12/2009	Shelton, IV	N/A	N/A
2010/0089970	12/2009	Smith et al.	N/A	N/A
2010/0127041	12/2009	Morgan et al.	N/A	N/A
2010/0133317	12/2009	Shelton, IV et al.	N/A	N/A
2010/0147921	12/2009	Olson	N/A	N/A
2010/0147922	12/2009	Olson	N/A	N/A
2010/0155453	12/2009	Bombard et al.	N/A	N/A
2010/0193566	12/2009	Scheib et al.	N/A	N/A
2010/0224668	12/2009	Fontayne et al.	N/A	N/A
2010/0249802	12/2009	May et al.	N/A	N/A
2010/0252611	12/2009	Ezzat et al.	N/A	N/A
2011/0006101	12/2010	Hall et al.	N/A	N/A
2011/0024477	12/2010	Hall	N/A	N/A
2011/0024478	12/2010	Shelton, IV	N/A	N/A
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2011/0114702	12/2010	Farascioni	N/A	N/A
2011/0121049	12/2010	Malinouskas et al.	N/A	N/A
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2011/0155787	12/2010	Baxter, III et al.	N/A	N/A
2011/0163146	12/2010	Ortiz et al.	N/A	N/A
2011/0163149	12/2010	Viola	N/A	N/A
2011/0192881	12/2010	Balbierz et al.	N/A	N/A
2011/0192882	12/2010	Hess et al.	N/A	N/A
2011/0192883	12/2010	Whitman et al.	N/A	N/A
2011/0204119	12/2010	McCuen	N/A	N/A
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2012/0241492	12/2011	Shelton, IV et al.	N/A	N/A
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2012/0241504	12/2011	Soltz et al.	N/A	N/A
2012/0248169	12/2011	Widenhouse et al.	N/A	N/A
2012/0286021	12/2011	Kostrzewski	N/A	N/A
2012/0286022	12/2011	Olson et al.	N/A	N/A
2012/0298722	12/2011	Hess et al.	N/A	N/A
2013/0008937	12/2012	Viola	N/A	N/A
2013/0012983	12/2012	Kleyman	N/A	N/A
2013/0015231	12/2012	Kostrzewski	N/A	N/A
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2013/0068816	12/2012	Mandakolathur Vasudevan et al.	N/A	N/A
2013/0068818	12/2012	Kasvikis	N/A	N/A
2013/0075447	12/2012	Weisenburgh, II et al.	N/A	N/A
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2013/0098964	12/2012	Smith et al.	N/A	N/A
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2013/0105545	12/2012	Burbank	N/A	N/A
2013/0105548	12/2012	Hodgkinson et al.	N/A	N/A
2013/0105552	12/2012	Weir et al.	N/A	N/A
2013/0105553	12/2012	Racenet et al.	N/A	N/A
2013/0112730	12/2012	Whitman et al.	N/A	N/A
2013/0119109	12/2012	Farascioni et al.	N/A	N/A
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2013/0146642	12/2012	Shelton, IV et al.	N/A	N/A
2013/0153636	12/2012	Shelton, IV et al.	N/A	N/A
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2014/0076955	12/2013	Lorenz	N/A	N/A

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2014/0166720	12/2013	Chowaniec et al.	N/A	N/A
2014/0166721	12/2013	Stevenson et al.	N/A	N/A
2014/0166724	12/2013	Schellin et al.	N/A	N/A
2014/0166725	12/2013	Schellin et al.	N/A	N/A
2014/0166726	12/2013	Schellin et al.	N/A	N/A
2014/0175146	12/2013	Knodel	N/A	N/A
2014/0175150	12/2013	Shelton, IV et al.	N/A	N/A
2014/0203062	12/2013	Viola	N/A	N/A
2014/0239036	12/2013	Zerkle et al.	N/A	N/A
2014/0239037	12/2013	Boudreaux et al.	N/A	N/A
2014/0239038	12/2013	Leimbach et al.	N/A	N/A
2014/0239040	12/2013	Fanelli et al.	N/A	N/A
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2014/0239043	12/2013	Simms et al.	N/A	N/A
2014/0239044	12/2013	Hoffman	N/A	N/A
2014/0239047	12/2013	Hodgkinson et al.	N/A	N/A
2014/0246471	12/2013	Jaworek	N/A	N/A
2014/0246472	12/2013	Kimsey et al.	N/A	N/A
2014/0246475	12/2013	Hall et al.	N/A	N/A
2014/0246478	12/2013	Baber et al.	N/A	N/A
2014/0252062	12/2013	Mozdzierz	N/A	N/A
2014/0252064	12/2013	Mozdzierz et al.	N/A	N/A
2014/0252065	12/2013	Hessler et al.	N/A	N/A
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2014/0263540	12/2013	Covach et al.	N/A	N/A
2014/0263541	12/2013	Leimbach et al.	N/A	N/A
2014/0263542	12/2013	Leimbach et al.	N/A	N/A
2014/0263544	12/2013	Ranucci et al.	N/A	N/A
2014/0263546	12/2013	Aranyi	N/A	N/A
2014/0263550	12/2013	Aranyi et al.	N/A	N/A
2014/0263552	12/2013	Hall et al.	N/A	N/A
2014/0263553	12/2013	Leimbach et al.	N/A	N/A
2014/0263554	12/2013	Leimbach et al.	N/A	N/A
2014/0263555	12/2013	Hufnagel et al.	N/A	N/A
2014/0263557	12/2013	Schaller	N/A	N/A
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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application is a continuation of U.S. patent application Ser. No. 17/329,711, filed May 25, 2021, now U.S. Pat. No. 11,771,423, the disclosure of which is incorporated by reference herein in its entirety.

FIELD

(1) This disclosure is directed to surgical devices and, more particularly, to powered surgical stapling devices.

BACKGROUND

(2) Various types of surgical devices used to endoscopically treat tissue are known in the art, and are commonly used, for example, for closure of tissue or organs in transection, resection, and anastomoses procedures, for occlusion of organs in thoracic and abdominal procedures, and for electrosurgically fusing or sealing tissue.

(3) One example of such a surgical device is a surgical stapling device. Typically, surgical stapling devices include a tool assembly having an anvil assembly and a cartridge assembly, and a drive assembly. Typically, the drive assembly includes a flexible drive beam and a clamp member that is supported on a distal end of the drive beam. The drive assembly is movable to advance the clamp member through the tool assembly to approximate the cartridge and anvil assemblies and to advance an actuation sled through the cartridge assembly to eject staples from the cartridge assembly.

(4) Surgical stapling devices can be manually actuated devices in which a clinician squeezes a trigger to actuate the stapling device, or powered stapling devices in which a clinician activates a motor within the stapling device to actuate the stapling device. Although powered stapling devices require less force to operate, difficulties may arise when the device loses power or components of the device break. In such instances, the device can remain clamped about tissue preventing removal of the device from a patient.

(5) A continuing need exists in the art for a powered stapling device that includes a drive assembly that can be manually retracted when power is lost or when the device is not operational.

SUMMARY

(6) A surgical device includes a powered handle assembly having a motor assembly, a rack, a spur gear, and a manual retract mechanism. The spur gear is movable from a position engaged with the motor assembly and the rack to a positioned disengaged from the motor assembly and engaged with the rack to facilitate manual retraction of the rack.

(7) One aspect of the disclosure is directed to a powered handle assembly for a surgical device that includes a housing, a gear casing, a motor assembly, a rack, a rotating shaft, and a spur gear. The housing defines a cavity. The gear casing is supported within the cavity of the housing and defines a longitudinal channel, a first cavity, and a second cavity that communicate with each other. The motor assembly includes an output shaft and a drive gear that is secured to the output shaft. The motor assembly is secured to the gear casing, and the drive gear is positioned within the second cavity of the gear casing. The rack is received within the longitudinal channel of the gear casing and is movable between retracted and advanced positions. The rotating shaft extends through the first cavity of the gear casing. The spur gear is coupled to the rotating shaft and is received within the first cavity of the gear casing. The spur gear is movable within the first cavity from a first position in which the spur gear is engaged with the drive gear and the rack to a second position in which the spur gear is disengaged from the drive gear and engaged with the rack.

(8) Other aspects of the disclosure are directed to a powered handle assembly for a surgical device that includes a housing, a motor assembly, a rack, a rotating shaft, and a spur gear. The housing defines a cavity. The motor assembly is supported within housing and includes an output shaft and

a drive gear that is secured to the output shaft. The rack is supported within the housing and is movable longitudinally between retracted and advanced positions. The rotating shaft is supported within the housing. The spur gear is coupled to the rotating shaft and received within the housing such that the spur gear is movable from a first position in which the spur gear is engaged with the drive gear and the rack to a second position in which the spur gear is disengaged from the drive gear and engaged with the rack.

(9) In aspects of the disclosure, the crank lever is coupled to the rotating shaft and is movable to move the spur gear from the first position to the second position.

(10) In some aspects of the disclosure, a biasing member is engaged with the spur gear and urges the spur gear towards the first position.

(11) In certain aspects of the disclosure, the rotating shaft includes a first portion and a second portion, wherein the first portion is rotatably fixed to the spur gear and the second portion receives the crank lever.

(12) In aspects of the disclosure, the housing defines an opening and includes a removable cover that is positioned over the opening such that the crank lever is accessible through the opening.

(13) In some aspects of the disclosure, the crank lever is movable along the second portion of the rotating shaft from a first position in which the rotating shaft can rotate independently of the crank lever to a second position in which the crank lever is rotatably fixed to the rotating shaft.

(14) In certain aspects of the disclosure, the crank lever includes a hub that defines a through bore having a rectangular portion and the second portion of the rotating shaft includes a rectangular portion that is received within the rectangular portion of the through bore when the crank lever is in its second position.

(15) In aspects of the disclosure, the handle assembly includes first and second C-clips, and the second portion of the rotating shaft defines spaced annular grooves that receive the first and second C-clips, respectively.

(16) In some aspects of the disclosure, the spur gear is received about the second portion of the rotating shaft atop the first C-clip within the first cavity of the gear casing and the second C-clip is positioned externally of the first cavity of the gear casing to secure the rotating shaft to the gear casing.

(17) Another aspect of the disclosure is directed to surgical stapling device that includes a powered handle assembly, an adapter assembly, and a tool assembly. The powered handle assembly includes a housing, a gear casing, a motor assembly, a rack, a rotating shaft, and a spur gear. The housing defines a cavity. The gear casing is supported within the cavity of the housing and defines a longitudinal channel, a first cavity, and a second cavity that communicate with each other. The motor assembly includes an output shaft and a drive gear secured to the output shaft. The motor assembly is secured to the gear casing, and the drive gear is positioned within the second cavity of the gear casing. The rack is received within the longitudinal channel of the gear casing and is movable between retracted and advanced positions. The rotating shaft extends through the first cavity of the gear casing. The spur gear is coupled to the rotating shaft and is received within the first cavity of the gear casing. The spur gear is movable within the first cavity from a first position in which the spur gear is engaged with the drive gear and the rack to a second position in which the spur gear is disengaged from the drive gear and engaged with the rack. The adapter assembly has a proximal portion coupled to the handle assembly and a distal portion. The adapter assembly includes a firing rod that is coupled to the rack and is movable between retracted and advanced positions in response to movement of the rack between its retracted and advanced positions. The tool assembly is supported on the distal portion of the adapter assembly.

(18) In aspects of the disclosure, the tool assembly includes an anvil and a cartridge assembly that are movable between open and clamped positions.

(19) In some aspects of the disclosure, the stapling device includes a drive assembly that is coupled to the firing rod and includes a working end having an I-beam configuration.

(20) In certain aspects of the disclosure, the working end of the drive assembly is movable in relation to the anvil and the cartridge assembly in response to movement of the firing rod between its retracted and advanced positions.

(21) Other aspects of the disclosure are directed to a powered handle assembly for a surgical device that includes a housing, a gear casing, a motor assembly, a drive screw, a drive nut, a connecting rod, and a spur gear. The housing defines a cavity. The gear casing is supported within the cavity of the housing and defines a channel. The motor assembly includes an output shaft and a drive gear secured to the output shaft. The motor assembly is secured to the gear casing, and the drive gear is positioned within the cavity of the gear casing. The drive screw is supported within the housing and is rotatable in response to activation of the motor assembly. The drive nut is supported on and movable along the drive screw between retracted and advanced positions. The connecting rod is coupled to the drive nut. The spur gear is movable within the channel of the gear casing from a first position in which the spur gear is engaged with the output shaft and the drive screw to a second position in which the spur gear is disengaged from the output shaft and engaged with the drive screw.

(22) Other aspects of the disclosure are directed to a powered handle assembly for a surgical device that includes a housing, a motor assembly, a drive screw, a drive nut, a connecting rod, a spur gear, a locking clip, and a pawl assembly. The housing defines a cavity. The motor assembly includes an output shaft and a drive gear secured to the output shaft. The motor assembly is positioned within the housing and the drive gear is positioned within the cavity of the gear casing. The drive screw is supported within the housing and is rotatable in response to activation of the motor assembly. The drive nut is supported on and movable along the drive screw between retracted and advanced positions. The connecting rod is coupled to the drive nut. The spur gear is movable within the channel of the gear casing from a first position in which the spur gear is engaged with the output shaft and the drive screw to a second position in which the spur gear is disengaged from the output shaft and engaged with the drive screw. The locking clip is movable from a first position retaining the spur gear in its first position to a second position allowing movement of the spur gear from its first position to its second position. The pawl assembly includes an annular body portion and a ratcheting pawl coupled to the body portion.

(23) In aspects of the disclosure, the handle assembly includes a biasing member that is positioned to urge the spur gear to the second position.

(24) In some aspects of the disclosure, the handle assembly includes a locking clip that is movable from a first position retaining the spur gear in its first position to a second position allowing movement of the spur gear from its first position to its second position.

(25) In certain aspects of the disclosure, the handle assembly includes a pawl assembly that includes a body portion and a ratcheting pawl coupled to the body portion.

(26) In aspects of the disclosure, the gear casing defines a window and the body portion of the pawl assembly is positioned about the gear casing adjacent the window such that the pawl assembly is movable from a first position in which the ratcheting pawl is spaced from the spur gear to a second position in which the ratcheting pawl is engaged with the spur gear.

(27) In some aspects of the disclosure, the connecting rod includes a proximal portion coupled to the drive nut and a distal portion coupled to a coupling member.

(28) In certain aspects of the disclosure, the coupling member is adapted to engage a firing rod of the surgical device.

(29) In aspects of the disclosure, the pawl assembly is coupled to the locking clip such that movement of the pawl assembly from its first position to its second position moves the locking clip from its first position to its second position.

(30) In some aspects of the disclosure, the gear casing defines spaced openings, and the locking clip includes legs that are received within the openings.

(31) Still other aspects of the disclosure are directed to a stapling device including a powered

handle assembly, an adapter assembly, and a tool assembly. The powered handle assembly includes a housing, a gear casing, a motor assembly, a drive screw, a drive nut, a connecting rod, and a spur gear. the housing defines a cavity. The gear casing is supported within the cavity of the housing and defines a channel. The motor assembly includes an output shaft and a drive gear secured to the output shaft. The motor assembly is secured to the gear casing, and the drive gear is positioned within the cavity of the gear casing. The drive screw is supported within the housing and is rotatable in response to activation of the motor assembly. The drive nut is supported on and movable along the drive screw between retracted and advanced positions. The connecting rod is coupled to the drive nut. The spur gear is movable within the channel of the gear casing from a first position in which the spur gear is engaged with the output shaft and the drive screw to a second position in which the spur gear is disengaged from the output shaft and engaged with the drive screw. The adapter assembly has a proximal portion coupled to the handle assembly and a distal portion. The adapter assembly includes a firing rod that is coupled to the drive nut and is movable between retracted and advanced positions in response to movement of the drive nut between its retracted and advanced positions. The tool assembly is supported on the distal portion of the adapter assembly.

(32) Other features of the disclosure will be appreciated from the following description.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various embodiments of the disclosed staple cartridge are described herein below with reference to the drawings, wherein:

(2) FIG. 1 is a side perspective view of a first version of a stapling device according to aspects of the disclosure with the stapling device in a non-articulated, unclamped position;

(3) FIG. 2 is a side cutaway view of a handle assembly of the stapling device shown in FIG. 1 with a cover of the handle assembly removed;

(4) FIG. 3 is a side perspective view of the handle assembly of the stapling device shown in FIG. 1 with a housing half-section removed;

(5) FIG. 4 is an exploded side perspective view of internal components of the handle assembly shown in FIG. 3;

(6) FIG. 5 is a side perspective view of a rotating shaft of the handle assembly shown in FIG. 3;

(7) FIG. 6 is a perspective view of a crank lever of the handle assembly shown in FIG. 3;

(8) FIG. 7 is a cross-sectional view taken along section line 7-7 of FIG. 2;

(9) FIG. 8 is a cross-sectional view taken through a portion of the handle assembly shown in FIG. 3 with the stapling device in an unclamped position;

(10) FIG. 9 is a cross-sectional view taken along section line 9-9 of FIG. 8;

(11) FIG. 10 is a side perspective view of a reload assembly of the stapling device shown in FIG. 1 in the clamped and fired position;

(12) FIG. 11 is a cross-sectional view taken along section line 11-11 of FIG. 10;

(13) FIG. 12 is a cross-sectional view through a portion of the handle assembly shown in FIG. 3 with the stapling device in the clamped and fired position;

(14) FIG. 13 is a side perspective view of a portion of the handle assembly shown in FIG. 3 as the stapling device is manually retracted from the clamped and fired position;

(15) FIG. 14 is a cross-sectional view taken along section line 14-14 of FIG. 13;

(16) FIG. 15 is a cross-sectional view taken along section line 15-15 of FIG. 14;

(17) FIG. 16 is a side perspective view of the handle assembly of the stapling device shown in FIG. 1 with the cover of the handle assembly removed as the stapling device is manually retracted;

(18) FIG. 17 is a cutaway, cross-sectional view taken through the handle assembly shown in FIG.

16 as the stapling device is manually retracted;

(19) FIG. **18** is an alternate version of the handle assembly of the stapling device shown in FIG. **1** with a housing of the handle assembly shown in phantom;

(20) FIG. **19** is an assembled view of the internal components of the handle assembly shown in FIG. **18**;

(21) FIG. **20** is an exploded side perspective view of internal components of the handle assembly shown in FIG. **19**;

(22) FIG. **21** is a side perspective view of a pawl assembly and locking clip of the handle assembly shown in FIG. **19**;

(23) FIG. **22** is a side perspective view from the proximal end of an interface between a motor assembly and a drive assembly of the handle assembly shown in FIG. **19** showing a manual retract mechanism with the locking clip in the locked position;

(24) FIG. **23** is a side perspective view from the distal end of the interface between the motor assembly and the drive assembly of the handle assembly shown in FIG. **20** with a gear casing removed showing the manual retract mechanism with the locking clip in the locked position;

(25) FIG. **24** is a cross-sectional view taken along section line **24-24** of FIG. **19**;

(26) FIG. **25** is a side cross-sectional view of the drive assembly of the handle assembly of the stapling device shown in FIG. **24** in the fired position;

(27) FIG. **26** is a side perspective view from the proximal end of the interface between the motor assembly and the drive assembly of the handle assembly shown in FIG. **19** showing the manual retract mechanism with the locking clip in the unlocked position;

(28) FIG. **26A** is a side cross-sectional view taken along section line **26A-26A** of FIG. **26** illustrating the manual retract mechanism in the unlocked position;

(29) FIG. **27** is a cross-sectional view taken along section line **27-27** of FIG. **26** illustrating the manual retract mechanism as the manual retract mechanism is operated to move the locking clip from the locked position to the unlocked position; and

(30) FIG. **28** is a cross-sectional view taken along section line **27-27** of FIG. **26** illustrating the manual retract mechanism as the manual retract mechanism is operated to retract the drive assembly.

DETAILED DESCRIPTION

(31) The disclosed surgical device will now be described in detail with reference to the drawings in which like reference numerals designate identical or corresponding elements in each of the several views. However, it is to be understood that the aspects of the disclosure are merely exemplary of the disclosure and may be embodied in various forms. Well-known functions or constructions are not described in detail to avoid obscuring the disclosure in unnecessary detail. Therefore, specific structural and functional details disclosed herein are not to be interpreted as limiting, but merely as a basis for the claims and as a representative basis for teaching one skilled in the art to variously employ the disclosure in virtually any appropriately detailed structure. In addition, directional terms such as front, rear, upper, lower, top, bottom, and similar terms are used to assist in understanding the description and are not intended to limit the disclosure.

(32) In this description, the term “proximal” is used generally to refer to that portion of the device that is closer to a clinician, while the term “distal” is used generally to refer to that portion of the device that is farther from the clinician. In addition, the term “endoscopic” is used generally to refer to endoscopic, laparoscopic, arthroscopic, and/or any other procedure conducted through a small diameter incision or cannula. Further, the term “clinician” is used generally to refer to medical personnel including doctors, nurses, surgeons, and support personnel.

(33) This disclosure is directed to a surgical device that includes a powered handle assembly having a motor assembly, a rack, a spur gear, and a manual retract mechanism. The spur gear is movable from a position engaged with the motor assembly and the rack to a position disengaged from the motor assembly and engaged with the rack to facilitate manual retraction of the rack.

(34) FIGS. 1 and 2 illustrate a surgical device shown generally as stapling device 10 which includes a handle assembly 12, an elongate body or adapter assembly 14, and a tool assembly 16. The handle assembly 12 includes a housing 18 that forms a stationary handle portion 18a, and articulation lever 19, and actuation buttons 20. The adapter assembly 14 defines a longitudinal axis "X" and includes a proximal portion 24 that is coupled to the handle assembly 12, and a distal portion 26 that supports the tool assembly 16. The tool assembly 16 is secured to the distal portion 26 of the adapter assembly 14 by a pivot member 28 that defines an axis "Y" that is transverse to the longitudinal axis "X". The articulation lever 19 is operatively coupled to the tool assembly 16 via an articulation linkage (not shown) such that manipulation of the articulation lever 19 causes articulation of the tool assembly 16 about the axis "Y" between an articulated position in which the tool assembly 16 is aligned with the longitudinal axis "Y" and non-articulated positions in which a longitudinal axis of the tool assembly and the longitudinal axis "X" define acute angles in response to manipulation of the articulation lever 19. The adapter assembly 14 is supported within a rotation knob 30 that is rotatably coupled to a distal portion of the handle assembly 12. The rotation knob 30 is manually rotatable about the longitudinal axis "X" to rotate the adapter assembly 14 and the tool assembly 16 about the longitudinal axis "X". The actuation buttons 20 control operation of the different functions of the stapling device 10 including clamping and firing of the stapling device 10.

(35) In aspects of the disclosure, the tool assembly 16 forms part of a reload assembly 32 that includes a proximal body portion 34 and the tool assembly 16. The proximal body portion 34 of the reload assembly 32 forms an extension of the adapter assembly 14 and includes a proximal end that is adapted to be releasably coupled to a distal end of the adapter assembly 14 and a distal end that supports the tool assembly 16 for articulation. In aspects of the disclosure, the tool assembly 16 can be fixedly coupled to a distal portion of the adapter assembly 14.

(36) In aspects of the disclosure, the housing 18 of the handle assembly 12 is formed from half-sections that are coupled together such as by welding of with screws to define a cavity 38 that receives internal components of the handle assembly 12 which are described in further detail below. The housing 18 defines an upper opening 40 that provides access to the internal components of the handle assembly 12. The upper opening 40 is enclosed by a cover 42 that is removably supported within the upper opening 40.

(37) FIGS. 3 and 4 illustrate the internal components of the handle assembly 12 which include a gear casing 44, a motor assembly 46, a rack 48, a firing rod 50, a manual retract mechanism 52 and intermediate spur gear 54, and a drive gear 56. The gear casing 44 is secured within the cavity 38 of the housing 18 using screws or the like and defines a first cavity 60 and a second cavity 62 that intersect with each other and a longitudinally extending channel 64. The first cavity 60 of the gear casing 44 receives the drive gear 56 and the second cavity 62 of the gear casing 44 receives the intermediate spur gear 54. The drive gear 56 and the intermediate spur gear each include gear teeth that mesh such that rotation of the drive gear 56 within the first cavity 60 causes corresponding rotation of the intermediate spur gear 54 within the second cavity 62. The rack 48 is received within the channel 64 of the gear casing 44 and includes gear teeth that mesh with the gear teeth of the intermediate spur gear 54. When the drive gear 56 is rotated to rotate the intermediate spur gear 54, engagement between the intermediate spur gear 56 and the rack 48 causes the rack 48 to move longitudinally through the channel 64 in the gear casing 44.

(38) The motor assembly 46 includes an output shaft 70 (FIG. 4) that is secured to the drive gear 56 and can be activated via the actuation buttons 20 (FIG. 1) to rotate the drive gear 56. In aspects of the disclosure, the motor assembly 46 is positioned within a portion of the cavity 38 of the housing 18 defined by the stationary handle portion 18a. The motor assembly 46 includes a mounting bracket 72 that is secured to the gear casing 44 with screws 74 such that the drive gear 56 is received within the second cavity 62 of the gear casing 44.

(39) FIGS. 4-6 illustrate the manual retract mechanism 52 which includes a rotating shaft 78, a

crank lever **80**, and a grip member **82**. The crank lever **80** includes a central hub portion **84** that defines a through bore **86** that receives the rotating shaft **78**. The through bore **86** includes a cylindrical portion **86a** and a rectangular portion **86b**. The crank lever **80** also includes a lever portion **88** that defines a slot **90** and forms a clevis **92**. The grip member **82** is supported within the clevis **92** by a pivot member **94** and is pivotable between a first position located within the slot **90** and a second position extending orthogonally from the lever portion **88**.

(40) The rotating shaft **78** includes a head portion **96** and a shaft portion **98** that extends downwardly from the head portion **96** as viewed in FIG. **4** through the through bore **86** in the crank lever **80**. The shaft portion **98** includes a first rectangular portion **100**, a cylindrical portion **102**, and a second rectangular portion **104**. The second rectangular portion **104** includes spaced annular grooves **106** and **108** that receive C-clips **110** and **112**, respectively. The intermediate spur gear **54** is received about the second rectangular portion **104** of the rotating shaft **78** atop the C-clip **112** within the first cavity **60** of the gear casing **44**. The C-clip **112** is positioned externally of the first cavity **60** and secures the rotating shaft **78** to the gear casing **44**. Although the rotating shaft **78** is shown to have first and second rectangular portions, it is envisioned that other shaft configurations are envisioned.

(41) The gear casing **44** defines first and second openings **114** and **116** that communicate with upper and lower ends of the first cavity **60** of the gear casing **44**. The cylindrical portion **102** of the rotating shaft **78** is received within the first opening **114** of the gear casing **44** and the second rectangular portion **104** of the rotating shaft **78** extends through the second opening **116** of the gear casing **44**. The C-clip **110** is secured to the lower end of the second rectangular portion **104** of the rotating shaft **78** adjacent a lower surface of the gear casing **44** and the C-clip **112** is secured to the second rectangular portion **104** of the rotating shaft **78** within the first cavity **60** of the gear casing **44**. The C-clips **110** and **112** secure the intermediate spur gear **54** to the rotating shaft **78** and secure the rotating shaft **78** to the gear casing **44**. The second rectangular portion **104** of the rotating shaft **78** is received within a rectangular bore in the intermediate spur gear **54** to rotatably fix the intermediate spur gear **54** to the rotating shaft **78**.

(42) The intermediate spur gear **54** is received within the first cavity **60** of the gear casing **44** and is movable within the first cavity between a first or lower position and a second or upper position. In the lower position (FIG. **8**), the teeth of the intermediate spur gear **54** simultaneously engage the teeth of the rack **48** and the teeth of the drive gear **56**. In the upper position, the teeth of the intermediate spur gear **54** engage only the teeth of the rack **48**.

(43) The manual retract mechanism **52** includes a biasing member **120** that is positioned between an upper surface of the intermediate spur gear **54** and the inner surface of the gear casing **44**. The biasing member **120** urges the intermediate spur gear **54** towards its lower position in which the intermediate spur gear **54** is engaged with both the drive gear **56** and the rack **48**. In some aspects of the disclosure, a washer **122** is positioned between an upper surface of the biasing member **120** and the inner surface of the gear casing **44**. In some aspects of the disclosure, the biasing member **120** includes a coil spring. It is envisioned that other types of biasing members could be incorporated into the stapling device **10** (FIG. **1**).

(44) The rack **44** includes a distal portion that is coupled to the firing rod **50**. In aspects of the disclosure, the proximal portion of the firing rod **50** is formed with a head **124** that has a diameter that is larger than a body of the firing rod **50** and the distal portion of the rack **48** defines a slot **128**. The head **124** of the firing rod **50** is received within the slot **128** of the rack **48** to couple the firing rod **50** to the rack **48** such that longitudinal movement of the rack **48** causes longitudinal movement of the firing rod **50**. The coupling of the head **124** and the slot **128** allows the firing rod **50** to rotate in relation to the rack **48**.

(45) In some aspects of the disclosure, the manual retract mechanism **52** includes a base member **130** that defines a circular bore **132**. The base member **130** is secured to an upper surface of the gear casing **44** and the hub portion **84** of the crank lever **80** is aligned within the circular bore **132**.

The crank lever **80** is movable from a lowered position to a raised position. In the lowered position, the hub portion **84** of the crank lever **80** is received within the circular bore **132** and in the raised position, the hub portion **84** of the crank lever **80** is positioned above the circular bore **132** of the base member **130**.

(46) The head **96** of the rotating shaft **78** of the manual retract mechanism **52** is received within the bore **86** of the crank lever **80**. When the crank lever **80** is in its lowered position, the cylindrical portion **102** of the rotating shaft **78** is received within the rectangular portion **86b** (FIG. **6**) of the through bore **86** of the crank lever **80**. In this position, the rotating shaft **78** can rotate within the through bore **86** of the crank lever **80** without causing rotation of the crank lever **80**. When the crank lever **80** is moved to its raised position, the first rectangular portion **100** of the rotating shaft **78** is received within the rectangular portion **86b** of the through bore **86** of the crank lever **80**. In this position, rotation of the crank lever **80** causes corresponding rotation of the rotating shaft **78**, and thus, rotation of the intermediate spur gear **54**. The head **96** of the rotating shaft **78** has a diameter that is larger than the width of the rectangular portion **86b** of the through bore **86** of the crank lever **80**. Thus, movement of the crank lever **80** to its raised position causes the rotating shaft **78** to its upper position in which the teeth of the intermediate spur gear **54** are only engaged with the teeth of the rack **48**.

(47) The handle assembly **12** (FIG. **1**) also includes one or more batteries **140** which are received within the cavity **38** of the housing **18** of the handle assembly **12**. The batteries **140** provide power to the motor assembly **46** via the actuation switches and control circuitry (not shown), e.g., a printed circuit board and one or more controllers, within the handle assembly **12** to control firing of the stapling device **10**.

(48) FIGS. **7-9** illustrate the handle assembly **12** of the stapling device **10** (FIG. **1**) with the stapling device in an unclamped position prior to firing of the stapling device. In this position, the rack **48** is in a retracted position within the channel **64** of the gear casing **44** and the intermediate spur gear **54** is in its lowered position and is engaged with the rack **48** and the drive gear **56**.

(49) FIGS. **10** and **11** illustrate the reload assembly **32** when the stapling device **10** is in a fired position. As described above, the reload assembly **32** includes the tool assembly **16** and the proximal body portion **34**. In aspects of the disclosure, the tool assembly **16** includes an anvil assembly **150** and a cartridge assembly **152**. The cartridge assembly **152** includes a staple cartridge **154** that supports a plurality of staples and pushers (not shown), and an actuation sled **156**. The proximal body portion **32** includes a drive assembly **158** that includes a flexible beam **160** and a working end **162**. The working end **162** of the drive assembly **158** has an I-beam configuration and is secured to a distal end portion of the flexible beam **160**. The flexible beam **160** has a proximal end portion that is releasably coupled to a distal portion of the firing rod **50**. When the firing rod **50** is moved from a retracted position to an advanced position, the drive assembly **158** moves from a retracted position to an advanced position to move the working end **162** of the drive assembly **158** through the tool assembly **16** to advance the actuation sled **156** through the tool assembly **16**. As the actuation sled **156** moves through the tool assembly **16**, the actuation sled **156** engages the pushers (not shown) to eject staples (not shown) from the staple cartridge **154** into the anvil assembly **150**. In the fired position, the working end **162** of the drive assembly **158** and the actuation sled **156** are in their advanced positions within the tool assembly and the tool assembly is in the clamped position clamped about tissue (not shown). For a more detailed description of the operation of the drive assembly **158** and its interaction with the tool assembly, see U.S. Pat. No. 8,132,706.

(50) FIG. **12** illustrates the handle assembly **12** of the stapling device **10** (FIG. **1**) as the stapling device **10** is fired. When the stapling device **10** is fired, the intermediate spur gear **54** (FIG. **8**) is engaged with the rack **48** and with the drive gear **56**. When the motor assembly **46** is activated, the drive gear **56** rotates the intermediate spur gear **54** to advance the rack **48** in the direction indicated by arrows "A". The rack **48** is coupled to the firing rod **50** such that advancement of the firing rod

50 advancement of the rack **48** causes advancement of the firing rod **50** in the direction of arrows “A” to advance the drive assembly **158** (FIG. **11**) within the tool assembly **16**.

(51) FIGS. **13-15** illustrate the manual retract mechanism **52** as it is readied for use. When the stapling device **10** loses power or gets damaged such that the motor assembly **46** cannot retract the drive assembly **158** to release tissue clamped between the anvil and cartridge assemblies **150** and **152** (FIG. **11**), the manual retract mechanism **52** can be operated to retract the drive assembly **158** (FIG. **11**). In order to access the manual retract mechanism **52**, the cover **42** (FIG. **2**) must be removed to uncover the upper opening **40** in the housing **18** of the handle assembly **12**. Once the cover is removed, the crank lever **80** is pulled upwardly in the direction of arrow “C” in FIG. **13** to move the crank lever **80** from its lowered position to its raised position. As the crank lever **80** is moved towards its raised position, the first rectangular portion **100** of the rotating shaft **78** is received in the rectangular portion **86b** of the through bore **86** in the crank lever **80**. Once the first rectangular portion **100** of the rotating shaft **78** is received in the rectangular portion **86b** of the through bore **86** in the crank lever **80**, continued movement of the crank lever **80** in the direction of arrow “C” will lift the rotating shaft **78** and the intermediate spur gear **54** in the direction of arrows “D” in FIG. **14** to their upper positions compressing the biasing member **120**. In the upper position, the intermediate spur gear **54** is disengaged from the drive gear **56** and is engaged only with the rack **48**. Once the crank lever **80** is in its raised position, the grip member **82** can be pivoted about the pivot member **94** to an operational position.

(52) FIGS. **16** and **17** illustrate the manual retract mechanism **52** as it is operated to retract the firing rod **50**. Once the crank lever **80** is moved to its raised position, the crank lever **80** can be rotated in the direction indicated by arrow “E” in FIG. **16** to rotate the rotating shaft **78** and the intermediate spur gear **54** in the direction indicated by arrow “F” in FIG. **17** to retract the rack **48**. More specifically, when the crank lever **80** is rotated, receipt of the first rectangular portion **100** of the rotating shaft **78** in the rectangular portion **86b** of the through bore **86** of the crank lever **80** rotatably fixes the crank lever **80** to the rotating shaft **78**. Thus, when the crank lever **80** rotates, the rotating shaft also rotates. The intermediate spur gear **54** is rotatably fixed to the rotating shaft **78** via receipt of the second rectangular portion **104** of the rotating shaft **78** in the rectangular bore **54a** of the intermediate spur gear **54** such that rotation of the rotating shaft **78** causes rotation of the intermediate spur gear **54**. In its upper position, the intermediate spur gear **54** is only engaged with the rack **48**, and as such, rotation of the intermediate spur gear **54** causes retract of the rack **48**.

(53) FIGS. **19-28** illustrate an alternate version of the handle assembly **12** (FIG. **18**) of the stapling device **10** shown generally as handle assembly **312**. The handle assembly **312** includes a housing **314** that is substantially similar to housing **18** (FIG. **1**) of stapling device **10** and will not be described in further detail herein. The housing **314** defines a cavity **316** that receives the internal components of the handle assembly **312**.

(54) FIGS. **18-20** illustrate the internal components of the handle assembly **312** which includes a motor assembly **318**, a drive assembly **320**, and a manual retract mechanism **322**. The motor assembly **318** is supported within the cavity **316** (FIG. **18**) of the housing **314** and includes an output shaft **324** that has a flat surface **324a**. In some aspects of the disclosure, the output shaft **324** has a D-shaped configuration although other configurations are envisioned.

(55) The drive assembly **320** is coupled to the output shaft **324** of the motor assembly **318** and includes a one-way spur gear **328**, a drive screw **330**, a drive nut **332**, connecting rods **334**, a coupling member **336**, a guide tube **338**, and a gear casing **340**. The gear casing **340** includes a mounting flange **342** and a cylindrical body **344**. The mounting flange **342** of the gear casing **340** defines bores **346** that receive screws **348**. The screws **348** are received in threaded bores **350** formed in a distal face of the motor assembly **318** to secure the gear casing **340** to the motor assembly **318**. The cylindrical body **344** of the gear casing **340** defines a cavity **352** and a window **354** that communicates with the cavity **352**. The cylindrical body **344** of the gear casing **340** defines two openings **356** (only one is shown) and two cutouts **358**. One of the openings **356** and

one of the cutouts **358** is positioned on each side of the window **354** in vertical alignment. The cavity **352** of the cylindrical body **344** of the gear casing **340** receives the one-way spur gear **328**. The distal portion of the cylindrical body **344** of the gear casing **340** supports a bearing **359**.

(56) The one-way spur gear **328** defines a central through bore **360** that receives a bearing **362**. In aspects of the disclosure, the central through bore **360** and the bearing **362** have corresponding non-circular configurations, e.g., D-shaped configurations, such that the bearing **362** is slidably received within the central through bore **360** of the one-way spur gear **328**. The corresponding configurations of the one-way spur gear **328** and the bearing **362** rotatably fix the components to each other. The bearing **362** also defines a central through bore **364** that has a non-circular configuration.

(57) The drive screw **330** includes a threaded outer surface **366**, a proximal extension **368**, and a distal extension **370**. The proximal extension **368** of the drive screw **330** extends through the bearing **359** within the gear casing **340** and is received and secured within the central through bore **364** of the bearing **362**. The distal extension **370** of the drive screw **330** is received within a bearing **372** that is supported within the housing **314** (FIG. 18) to rotatably support the drive screw **330** within the housing **314**.

(58) When the motor assembly **318** is activated to rotate the output shaft **324**, rotation of the output shaft **324**, when engaged with the one-way spur gear **328**, causes corresponding rotation of the one-way spur gear **328**. As described above, the one-way spur gear **328** is rotatably fixed to the bearing **362** which is secured to and rotatably fixed to the drive screw **330**. As such, rotation of the one-way spur gear **328** causes corresponding rotation of the drive screw **330**.

(59) The drive nut **332** includes a threaded bore **374** that receives and is threadably engaged with the threaded outer surface **366** of the drive screw **330**. The drive nut **332** is coupled to a proximal portion of the connecting rods **334**. In aspects of the disclosure, the drive nut **332** includes protrusions **378** that are received within openings **380** formed in the proximal portions of the connecting rods **334** to connect the drive nut **332** to the connecting rods **334**. The connecting rods **334** extend distally from the drive nut **332** and include distal portions that are connected to the coupling member **336**. In aspects of the disclosure, the coupling member **336** includes protrusions **384** that are received within openings **386** formed in the distal portions of the connecting rods **334** to connect the coupling member **336** to the connecting rods **334**. The drive nut **332** and the connecting rods **334** are received within the guide tube **338**.

(60) When the drive screw **330** is rotated, engagement between the outer threaded surface **366** of the drive screw **330** and the inner threaded bore **374** of the drive nut **332** causes the drive nut **332** to translate longitudinally along the drive screw **330** within the guide tube **338**. The drive nut **332** is connected to the connecting rods **334** such that longitudinal translation of the drive nut **332** along the drive screw **330** causes the connecting rods **334** to move longitudinally within the guide tube to advance to coupling member **336**.

(61) The coupling member **336** is coupled to a firing rod **382** such that longitudinal movement of the coupling member **336** causes longitudinal movement of the firing rod **382**. In aspects of the disclosure, the firing rod **382** includes a head portion **384** and an elongate body **386**. The head portion **384** has a diameter that is greater than a diameter of the elongate body **386**. The coupling member **336** defines a slot **390** that has a width that is greater than the diameter of the elongate body **386** but less than the diameter of the head portion **384**. The elongate body **386** of the firing rod **382** is received through the slot **390** in the coupling member **336** to axially fix the firing rod **382** to the coupling member **336** while allowing relative rotation of the firing rod **382** and the coupling member **336**.

(62) The one-way spur gear **328** is movably positioned within the cavity **352** of the cylindrical body **344** of the gear casing **340** between a retracted position (FIG. 24) and an advanced position (FIG. 26A). In the advanced position, the one-way spur gear is engaged with the proximal extension **368** of the drive screw **330** and the output shaft **324** of the motor assembly **318** such that

rotation of the output shaft **324** of the motor assembly **324** causes rotation of the drive screw **330**. In the advanced position, the one-way spur gear **328** is disengaged from the output shaft **324** of the motor assembly **318** but still engaged with the drive screw **330**. A biasing member **396**, e.g., a coil spring, is positioned between the distal surface of the motor assembly **318** and a proximal surface of the one-way spur gear **328** to urge the one-way spur gear **328** towards the advanced position. In aspects of the disclosure, the proximal surface of the one-way spur gear **328** defines a recess **328a** (FIG. **26A**) that receives the biasing member **396**.

(63) FIGS. **20** and **21** illustrate the manual retract mechanism **322** includes a pawl assembly **410** and a locking clip **412**. The pawl assembly **410** includes a handle **414** and a body portion **416**. In aspects of the disclosure, the body portion has an oval or annular configuration and supports a ratcheting pawl **418** that is pivotably secured to an upper portion of the body portion **416** by a pivot member **420**. The ratcheting pawl **418** extends downwardly into a circular opening defined by the body portion **416**. The body portion **416** is received about the gear casing **340** with the ratcheting pawl **418** positioned over the window **354** in the gear casing **340** above the one-way spur gear **328**. A lower portion of the body portion **416** defines a circular slot **422**.

(64) The locking clip **412** has a rectangular shape and includes a base portion **426** and spaced legs **428** that extend upwardly from the base portion **426**. Each of the legs **428** of the locking clip **412** includes a stepped inner surface **430** that includes a first surface **430a** and a second surface **430b**. The first surfaces **430a** of the legs **428** define a first width $X1$ and the second surfaces **430b** define a second width $X2$ that is greater than the first width $X1$. Each of the legs **428** is received through one of the openings **356** and cutouts **358** of the gear casing **340** such that the stepped inner surfaces **430** of the legs **428** of the locking clip **412** are positioned within the cavity **352** of the gear casing **340**. The locking clip **322** is movable from a first position (FIG. **23**) in which the first surfaces **430a** of the legs **428** of the locking clip **322** are aligned with the one-way spur gear **328** and a second position in which the second surfaces **430b** of the legs **428** of the locking clip **322** are spaced from the one-way spur gear **328**. The width $X1$ between the first surfaces **430a** of the legs **428** of the locking clip **322** is such to prevent movement of the one-way spur gear **322** to its advanced position, whereas the width $X2$ between the second surfaces **430b** of the legs **428** of the locking clip **322** allows movement of the one-way spur gear **322** to the advanced position.

(65) The base portion **426** of the locking clip **322** includes a protrusion **434** that is received within the circular slot **422** in the body portion **416** of the pawl assembly **410**. Receipt of the protrusion **434** couples the pawl assembly **410** to the locking clip **322** to retain the pawl assembly **410** in a stable position about the gear casing **340**. In aspects of the disclosure, the protrusion **434** has an enlarged head and the circular slot includes overhanging ledges that retain the enlarged head of the protrusion **434** within the circular slot **422**. The protrusion **434** is configured to slide within the circular slot **422** as described in further detail below.

(66) FIGS. **22-24** illustrate the handle assembly **12** (FIG. **18**) in a pre-fired position with the pawl assembly **322** positioned about the gear casing **340** and the ratcheting pawl **418** positioned above the window **354** in the gear casing **340**. When the handle assembly **12** is assembled, the one-way spur gear **328** is pressed proximally towards the motor assembly **318** to compress the biasing member **396** and position the one-way spur gear **328** in its retracted position. After the one-way spur gear **328** is in its retracted position, the legs **428** of the locking clip **412** are inserted from a side of the gear casing **340** opposite to the ratcheting pawl **418** into the openings **356** and cutouts **358** formed in the gear casing **340** to its first position. In the first position of the locking clip **412**, the first surfaces **430a** of the legs **428** of the locking clip **412** engage a distal face of the one-way spur gear **328** to retain the one-way spur gear **328** in its retracted position against the urging of the biasing member **396**. In its retracted position, the one-way spur gear **328** is engaged with both the output shaft **324** of the motor assembly **318** and the one-way spur gear **328**. When the locking clip **412** is in its first position, the protrusion **434** on the locking clip **412** is pressed into the circular slot **422** on the body portion **416** of the pawl assembly **322** to couple the pawl assembly **322** to the

locking clip **412** (FIG. **24**).

(67) In the pre-fired position of the handle assembly **12** (FIG. **18**), the drive nut **322** is positioned near the proximal end of the drive screw **330** and the coupling member **336** is positioned adjacent the distal end of the guide tube **338** such that the connecting rods **334** are in retracted positions and the firing rod **382** is in its retracted position.

(68) FIG. **25** illustrates the handle assembly **12** (FIG. **1**) in a fired position. When the stapling device **10** (FIG. **1**) is fired by pressing the actuation buttons **20** (FIG. **1**), the motor assembly **318** is activated to rotate the output shaft **324**. Rotation of the output shaft **324** causes corresponding rotation of the one-way spur gear **328** and the drive screw **330** to advance the drive nut **322** along the drive screw **330** in the direction of arrow “J”. As the drive screw **330** advances the drive nut **322**, the drive nut **322** advances the connecting rods **334** to advance the firing rod **382** in the direction of arrow “K” and actuate the tool assembly **16** (FIG. **1**) as described above regarding stapling device **10** (FIG. **1**).

(69) When the tool assembly **16** is in the clamped and fired position (FIG. **11**) and the powered stapling device **10** (FIG. **1**) becomes inoperable and cannot be unclamped using the motor assembly **318**, the manual retract mechanism **322** allows the tool assembly to be manually unclamped. FIGS. **26-28** illustrate operation of the manual retract mechanism **322**. In order to operate the manual retract mechanism **322**, the pawl assembly **322** is pressed downwardly in the direction of arrows “L” in FIGS. **26-27**. When the pawl assembly **322** is pressed downwardly, the locking clip **412**, which is coupled to the pawl assembly **322** by the protrusion **434**, is moved from its first position to its second position. In its second position, the locking clip **412** disengages from the one-way spur gear **328** such that the biasing member **396** moves the one-way spur gear **328** from its retracted position to its advanced position. In its advanced position, the one-way spur gear **328** is disengaged from the output shaft **324** of the motor assembly **318**. When the pawl assembly **322** is pressed downwardly, the ratcheting pawl **418** moves through the window **354** of the gear housing **340** into engagement with the one-way spur gear **328**.

(70) After the one-way spur gear **328** is in its advanced position, the handle **414** of the pawl assembly **322** can be rotated in the direction of arrow “M” in FIG. **28** to rotate the one-way spur gear **328** and the drive screw **330** to retract the firing rod **382**. As the pawl assembly **322** is rotated, the protrusion **434** moves within the circular slot **422** of the pawl assembly **322**.

(71) Persons skilled in the art will understand that the devices and methods specifically described herein and illustrated in the accompanying drawings are non-limiting exemplary embodiments. It is envisioned that the elements and features illustrated or described in connection with one exemplary embodiment may be combined with the elements and features of another without departing from the scope of the disclosure. As well, one skilled in the art will appreciate further features and advantages of the disclosure based on the above-described embodiments. Accordingly, the disclosure is not to be limited by what has been particularly shown and described, except as indicated by the appended claims.

Claims

1. A powered handle assembly for a surgical device comprising: a housing defining a cavity; a motor assembly including an output shaft and a drive gear secured to the output shaft, the motor assembly supported within the housing; a rack received within the housing and movable between a retracted position and an advanced position; a rotating shaft supported within the housing, the rotating shaft including a first portion and a second portion; a spur gear coupled to the rotating shaft such that rotation of the rotating shaft causes rotation of the spur gear, wherein the first portion of the rotating shaft is rotatably fixed to the spur gear, wherein the rotating shaft is movable from a first shaft position to a second shaft position to move the spur gear from a first gear position in which the spur gear is engaged with the drive gear and the rack to a second gear position in

which the spur gear is disengaged from the drive gear and engaged with the rack; and a crank lever coupled to the rotating shaft, wherein the second portion of the rotating shaft is engaged with the crank lever, and wherein the crank lever is movable to move the spur gear from the first gear position to the second gear position.

2. The powered handle assembly of claim 1, further including a biasing member engaged with the spur gear, the biasing member positioned to urge the spur gear towards the first gear position.

3. The powered handle assembly of claim 1, wherein the housing defines an opening and includes a removable cover positioned over the opening, the crank lever accessible through the opening.

4. The powered handle assembly of claim 3, wherein the crank lever is movable along the second portion of the rotating shaft from a first lever position in which the rotating shaft can rotate independently of the crank lever to a second lever position in which the crank lever is rotatably fixed to the rotating shaft.

5. The powered handle assembly of claim 4, wherein the crank lever includes a hub that defines a through bore having a rectangular portion, and the second portion of the rotating shaft includes a rectangular portion that is received within the rectangular portion of the through bore when the crank lever is in the second lever position.

6. The powered handle assembly of claim 1, further including first and second C-clips, the second portion of the rotating shaft defining spaced annular grooves that receive the first and second C-clips.

7. A powered handle assembly for a surgical device comprising: a housing defining a cavity; a motor assembly including an output shaft and a drive gear secured to the output shaft, the motor assembly supported within the housing; a rack received within the housing and movable between retracted and advanced positions; a rotating shaft supported within the housing and accessible through the housing, the rotating shaft including a first portion and a second portion; a spur gear coupled to the rotating shaft such that rotation of the rotating shaft causes rotation of the spur gear, wherein the first portion of the rotating shaft is rotatably fixed to the spur gear, wherein the spur gear is received within the housing, the spur gear movable from a first gear position in which the spur gear is engaged with the drive gear and the rack to a second gear position in which the spur gear is disengaged from the drive gear and engaged with the rack; and a crank lever coupled to the rotating shaft, wherein the second portion of the rotating shaft is engaged with the crank lever, and wherein the crank lever is movable to move the spur gear from the first gear position to the second gear position.

8. The powered handle assembly of claim 7, further including a biasing member engaged with the spur gear, the biasing member positioned to urge the spur gear towards the first gear position.

9. The powered handle assembly of claim 7, wherein the housing defines an opening and includes a removable cover positioned over the opening, the crank lever accessible through the opening.

10. The powered handle assembly of claim 9, wherein the crank lever is movable along the second portion of the rotating shaft from a first lever position in which the rotating shaft can rotate independently of the crank lever to a second lever position in which the crank lever is rotatably fixed to the rotating shaft.

11. The powered handle assembly of claim 10, wherein the crank lever includes a hub that defines a through bore having a rectangular portion, and the second portion of the rotating shaft includes a rectangular portion that is received within the rectangular portion of the through bore when the crank lever is in the second lever position.

12. The powered handle assembly of claim 7, further including first and second C-clips, the second portion of the rotating shaft defining spaced annular grooves that receive the first and second C-clips.

13. A powered handle assembly for a surgical device comprising: a motor assembly including an output shaft and a drive gear secured to the output shaft; a rack movable in relation to the motor assembly between retracted and advanced positions; a rotating shaft including a first portion and a

second portion; a spur gear coupled to the rotating shaft, wherein the first portion of the rotating shaft is rotatably fixed to the spur gear, wherein the rotating shaft is movable from a first shaft position to a second shaft position to move the spur gear from a first gear position in which the spur gear is engaged with the drive gear and the rack to a second gear position in which the spur gear is disengaged from the drive gear and engaged with the rack; and a crank lever coupled to the rotating shaft, wherein the second portion of the rotating shaft is engaged with the crank lever, and wherein the crank lever is movable to move the spur gear from the first gear position to the second gear position.

14. The powered handle assembly of claim 13, further including a biasing member engaged with the spur gear, the biasing member positioned to urge the spur gear towards the first gear position.
